

```
In [2]: 1 import array
        2 import cmath
        3 import collections
        4 import copy
        5 import csv
        6 import datetime
        7 import decimal
        8 import functools
        9 import importlib
       10 import logging
       11 import math
       12 import pprint
       13 import random
       14 import re
       15 import statistics
       16 import string
       17 import sys
       18 import time
       19 import timeit
       20
       21 import pandas
```

```
In [2]: 1 num1 = 10_00_000
        2 print(num1)
```

```
1000000
```

```
In [3]: 1 print(format(0.1, ".25f"))
```

```
0.10000000000000000055511151
```

```
In [4]: 1 print(format(0.125, ".25f"))
```

```
0.12500000000000000000000000
```

```
In [5]: 1 print(0.1 + 0.1 + 0.1 == 0.3)
```

```
False
```

```
In [6]: 1 print(format(0.3, ".25f"))
```

```
0.2999999999999999888977698
```

```
In [7]: 1 print(format(0.2, ".25f"))
```

```
0.2000000000000000111022302
```

```
In [8]: 1 print(abs((0.1 + 0.1 + 0.1) - 0.3) < 0.001)
```

```
True
```

```
In [9]: 1 print((0.125).as_integer_ratio())
```

(1, 8)

```
In [10]: 1 print((10).      (25))
```

35

```
In [11]: 1 a = float(10)
          2 print(a)
```

10.0

```
In [12]: 1 c = (-4) ** 0.5
          2 print(c)
```

(1.2246467991473532e-16+2j)

```
In [13]: 1 print(type(c))
```

<class 'complex'>

```
In [14]: 1 print(c.real)
```

1.2246467991473532e-16

```
In [15]: 1 print(c.imag)
```

2.0

```
In [16]: 1 c = 10 + 2j
          2 print(type(c))
```

<class 'complex'>

```
In [17]: 1 print(c)
```

(10+2j)

```
In [18]: 1 try:
          2     print("2" < 2)
          3 except TypeError as ex:
          4     print(type(ex).      )
```

TypeError

```
In [19]: 1 print("2" == 2)
```

False

```
In [20]: 1 a = 0
          2 b = 1
          3 c = 500
          4 d = 0
          5 e = 1
          6 f = 500
          7 print(id(a), id(b), id(c), id(d), id(e), id(f))
```

11370064 11370096 128573173327504 11370064 11370096 128573173326608

```
In [21]: 1 print(0.1 * 3 - 0.3 < abs(1e-6))
```

```
True
```

```
In [22]: 1 a = 257
        2 print(a is 257)
```

```
False
```

```
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
```

```
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
```

```
/tmp/ipykernel_18021/1958631963.py:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
      print(a is 257)
```

```
In [23]: 1 a = 10
        2 b = 0
        3 print(b != 0 and a / b)
```

```
False
```

```
In [24]: 1 a = 1e-6
        2 b = 2e-6
        3 tol = 1e-7
        4 print(abs(a - b) <= tol)
```

```
False
```

```
In [25]: 1 elapsed = 7835
        2 hours = elapsed // 3600
        3 minutes = (elapsed - (hours * 3600)) // 60
        4 seconds = elapsed - ((hours * 3600) + (minutes * 60))
        5 print(hours, minutes, seconds)
```

```
2 10 35
```

```
In [26]: 1 elapsed = 7835
        2 hours = elapsed // (60 * 60)
        3 minutes = (elapsed % (60 * 60)) // 60
        4 seconds = (elapsed % (60 * 60)) % 60
        5 print(hours, minutes, seconds)
```

```
2 10 35
```

```
In [27]: 1 a = None
        2 if a == None:
        3     a = "N/A"
        4 print(a)
```

```
N/A
```

```
In [28]: 1 a = None
        2 a = "N/A" if a is None else a
        3 print(a)
```

```
N/A
```

In [29]:

```
1 score = 700
2 rating = "Invalid"
3 if score < 0:
4     rating = "Negative"
5 elif score < 580:
6     rating = "Poor"
7 elif score < 670:
8     rating = "Fair"
9 elif score < 740:
10    rating = "Good"
11 elif score < 800:
12    rating = "Very Good"
13 elif score < 850:
14    rating = "Excellent"
15 print(rating)
```

Good

In [30]:

```
1 elapsed = 1000000
2 magnitude = "Invalid"
3 if elapsed < 60:
4     magnitude = "seconds"
5 elif elapsed < (60 * 60):
6     magnitude = "minutes"
7 elif elapsed < (60 * 60 * 24):
8     magnitude = "hours"
9 elif elapsed < (60 * 60 * 24 * 7):
10    magnitude = "days"
11 elif elapsed > (60 * 60 * 24 * 7):
12    magnitude = "weeks"
13 print(magnitude)
```

weeks

In [31]:

```
1 t = [1, 2], 3, 4
2 l = list(t)
3 print(l)
```

[[1, 2], 3, 4]

In [32]:

```
1 l[0][1] = 20
2 print(l)
```

[[1, 20], 3, 4]

In [33]:

```
1 print(t)
```

([1, 20], 3, 4)

In [34]:

```
1 for i in t:
2     print(id(i))
```

```
128573172499456
11370160
11370192
```

```
In [35]: 1 for i in l:
          2     print(id(i))
```

```
128573172499456
11370160
11370192
```

```
In [36]: 1 for i, j in zip(l, t):
          2     print(i is j)
```

```
True
True
True
```

```
In [37]: 1 s = "ABCDEFGHIIJ"
          2 print(s[3:6:-1])
```

```
In [38]: 1 print(s[3:5 + 1])
```

```
DEF
```

```
In [39]: 1 m1 = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m2 = m1[0:2]
          3 print(m1[0] is m2[1])
```

```
False
```

```
In [40]: 1 m1[0] = "Python"
          2 print(m1)
```

```
['Python', [0, 0, 0], [0, 0, 0]]
```

```
In [41]: 1 print(m2)
```

```
[[0, 0, 0], [0, 0, 0]]
```

```
In [42]: 1 m1[1][0] = 100
```

```
In [43]: 1 print(m1)
```

```
['Python', [100, 0, 0], [0, 0, 0]]
```

```
In [44]: 1 print(m2)
```

```
[[0, 0, 0], [100, 0, 0]]
```

```
In [45]: 1 print("abcdef"[-1:-4])
```

```
In [46]: 1 print("abcdef"[1:4:-1])
```

```
In [47]: 1 print("racecar" == "racecar"[::-1])
```

```
True
```

```
In [48]: 1 list1 = [0, 1, 2, 3, 4]
        2 list1[0:3] = "abc"
        3 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [49]: 1 list1[0:3] = ["a", "b", "c"]
        2 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [50]: 1 list1[0:3] = ["a", "b"]
        2 print(list1)
```

```
['a', 'b', 3, 4]
```

```
In [51]: 1 del list1[2]
        2 print(list1)
```

```
['a', 'b', 4]
```

```
In [52]: 1 del list1[0:2]
        2 print(list1)
```

```
[4]
```

```
In [53]: 1 list1.extend("ABC")
        2 print(list1)
```

```
[4, 'A', 'B', 'C']
```

```
In [54]: 1 list1.insert(0, "YO!")
        2 print(list1)
```

```
['YO!', 4, 'A', 'B', 'C']
```

```
In [55]: 1 list1[:-3:-1] = "cb"
        2 print(list1)
```

```
['YO!', 4, 'A', 'b', 'c']
```

```
In [56]: 1 l = []
        2 print(timeit.timeit("l.append(1)", globals = globals(), number = 100_000))
```

```
0.002112314999976661
```

```
In [57]: 1 print(len(l))
```

```
100000
```

```
In [58]: 1 list2 = []
        2 print(timeit.timeit("list2.insert(0, 1)", globals = globals(), number =
```

```
100_000))
```

```
1.5377864980000027
```

```
In [59]: 1 print(len(list2))
```

```
100000
```

```
In [60]: 1 print(1.249788 / 0.0018043)
```

```
692.6719503408524
```

```
In [61]: 1 list2 = []  
2 print(timeit.timeit("list2.insert(len(list2), 1)", globals = globals(), number  
= 100_000))
```

```
0.008461738999983481
```

```
In [62]: 1 print(0.0057634 / 0.0018043)
```

```
3.194258161059691
```

```
In [63]: 1 l1 = [[0, 1], 2, "3"]  
2 l2 = l1[:]  
3 for i in l1:  
4     print(id(i))
```

```
128573172569024
```

```
11370128
```

```
11434576
```

```
In [64]: 1 for i in l2:  
2     print(id(i))
```

```
128573172569024
```

```
11370128
```

```
11434576
```

```
In [65]: 1 l2[0][0] = "0"  
2 for i in l1:  
3     print(id(i))
```

```
128573172569024
```

```
11370128
```

```
11434576
```

```
In [66]: 1 for i in l2:  
2     print(id(i))
```

```
128573172569024
```

```
11370128
```

```
11434576
```

```
In [67]: 1 for i in l1:  
2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [68]: 1 for i in l2:  
        2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [69]: 1 l2[1] = "22"  
        2 for i in l1:  
        3     print(id(i))
```

```
128573172569024  
11370128  
11434576
```

```
In [70]: 1 for i in l2:  
        2     print(id(i))
```

```
128573172569024  
128574645937104  
11434576
```

```
In [71]: 1 for i in l1:  
        2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [72]: 1 for i in l2:  
        2     print(i)
```

```
['0', 1]  
22  
3
```

```
In [73]: 1 l1 = [['0', 1], 2, "3"]  
        2 l2 = l1[:]  
        3 l2[0] = ["0", 1]  
        4 for i in l1:  
        5     print(id(i))
```

```
128573172599040  
11370128  
11434576
```

```
In [74]: 1 for i in l2:  
        2     print(id(i))
```

```
128573172596864  
11370128  
11434576
```

```
In [75]: 1 for i in l1:
        2     print(i)
```

```
[0, 1]
2
3
```

```
In [76]: 1 for i in l2:
        2     print(i)
```

```
['0', 1]
2
3
```

```
In [77]: 1 l1 = [[0, 1], 2, "3"]
        2 l2 = l1[:]
        3 print(id(l1), id(l2))
```

```
128573172599104 128573172598912
```

```
In [78]: 1 for i, j in zip(l1, l2):
        2     print(id(i), id(j))
```

```
128573172598848 128573172598848
11370128 11370128
11434576 11434576
```

```
In [79]: 1 for i, j in zip(l1[0], l2[0]):
        2     print(id(i), id(j))
```

```
11370064 11370064
11370096 11370096
```

```
In [80]: 1 l1 = [[0, 1], 2, "3"]
        2 l2 = l1.copy()
        3 print(id(l1), id(l2))
```

```
128573172597248 128573172597312
```

```
In [81]: 1 for i, j in zip(l1, l2):
        2     print(id(i), id(j))
```

```
128573172601472 128573172601472
11370128 11370128
11434576 11434576
```

```
In [82]: 1 for i, j in zip(l1[0], l2[0]):
        2     print(id(i), id(j))
```

```
11370064 11370064
11370096 11370096
```

```
In [83]: 1 l1 = [[0, 1], 2, "3"]
        2 l2 = copy.deepcopy(l1)
        3 print(id(l1), id(l2))
```

```
128574051537216 128573172602368
```

```
In [84]: 1 for i, j in zip(l1, l2):  
        2     print(id(i), id(j))
```

```
128573172600640 128573172437184  
11370128 11370128  
11434576 11434576
```

```
In [85]: 1 for i, j in zip(l1[0], l2[0]):  
        2     print(id(i), id(j))
```

```
11370064 11370064  
11370096 11370096
```

```
In [86]: 1 t1 = (0, "1", "23")  
        2 t2 = t1[:]  
        3 print(t1 is t2)
```

```
True
```

```
In [87]: 1 a, b, c, d = "abcd"  
        2 print(a, b, c, d)
```

```
a b c d
```

```
In [88]: 1 s = "FfEeDdCcBbAa"  
        2 lowerLetters = s[::-2]  
        3 upperLetters = s[-2::-2]  
        4 print(lowerLetters)
```

```
abcdef
```

```
In [89]: 1 print(upperLetters)
```

```
ABCDEF
```

```
In [90]: 1 t1 = 1, 2, 3, 4, 5, 6  
        2 t2 = 7, 8, 9, 10  
        3 t3 = 11, 12, 13, 14, 15, 16, 17  
        4 t = t1 + t2 + t3  
        5 l = list(t)  
        6 l[::2] = [0] * len(l[::2])  
        7 print(tuple(l))
```

```
(0, 2, 0, 4, 0, 6, 0, 8, 0, 10, 0, 12, 0, 14, 0, 16, 0)
```

```
In [91]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]  
        2 m[0][0] = 1  
        3 m[1][1] = 1  
        4 m[2][2] = 1  
        5 print(m)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [92]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
2 m1 = copy.deepcopy(m)
3 m1[0][0] = 1
4 m1[1][1] = 1
5 m1[2][2] = 1
6 print(m1)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [93]: 1 print(m)
2 data = [(100, "USD", "EUR", 0.83), (100, "USD", "CAD", 1.27), (100, "CAD",
3 "EUR", 0.65)]
4 USDAmountInEUR = data[0][0] * data[0][3]
5 print(str(data[0][0]) + " " + data[0][1] + " = " + str(USDAmountInEUR) + " " +
str(data[0][2]))
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
100 USD = 83.0 EUR
```

```
In [94]: 1 USDAmountInCAD = data[1][0] * data[1][3]
2 print(str(data[1][0]) + " " + data[1][1] + " = " + str(USDAmountInCAD) + " " +
str(data[1][2]))
```

```
100 USD = 127.0 CAD
```

```
In [95]: 1 CADAmountInEUR = data[2][0] * data[2][3]
2 print(str(data[2][0]) + " " + data[2][1] + " = " + str(CADAmountInEUR) + " " +
str(data[2][2]))
```

```
100 CAD = 65.0 EUR
```

```
In [96]: 1 print(ord("A"))
```

```
65
```

```
In [97]: 1 print("\N{SNAKE}")
```

```
🐍
```

```
In [98]: 1 print(ord("\N{SNAKE}"))
```

```
128013
```

```
In [99]: 1 print(hex(ord("\N{SNAKE}"))) 
```

```
0x1f40d
```

```
In [100]: 1 print("\u03b1")
```

```
α
```

```
In [101]: 1 print("\U0001f40d")
```

```
🐍
```

```
In [102]: 1 print(int("1f40d", 16))
```

```
128013
```

```
In [103]: 1 print("\N{SNAKE} is a snake")
```

```
🐍 is a snake
```

```
In [104]: 1 print(int("7ff", 16))
```

```
2047
```

```
In [105]: 1 print(int("ffff", 16))
```

```
65535
```

```
In [106]: 1 print(int("10ffff", 16))
```

```
1114111
```

```
In [107]: 1 print(bin(2047))
```

```
0b1111111111
```

```
In [108]: 1 print(len(bin(2047)))
```

```
13
```

```
In [109]: 1 print(bin(65535))
```

```
0b1111111111111111
```

```
In [110]: 1 print(len(bin(65535)))
```

```
18
```

```
In [111]: 1 print(bin(1114111))
```

```
0b10000111111111111111
```

```
In [112]: 1 print(len(bin(1114111)))
```

```
23
```

```
In [113]: 1 print(bin(128013))
```

```
0b111111010000001101
```

```
In [114]: 1 print("\n{snake}")
```

```
{snake}
```

```
In [115]: 1 a = True  
2 b = True  
3 print(a is b)
```

```
True
```

```
In [116]: 1 a = False
          2 b = False
          3 print(a is b)
```

```
True
```

```
In [117]: 1 a = True
          2 print(id(a))
```

```
11120192
```

```
In [118]: 1 a = False
          2 print(id(a))
```

```
11045056
```

```
In [119]: 1 b = False
          2 print(id(b))
```

```
11045056
```

```
In [120]: 1 ?
```

**Docstring:**

Return the lowest index in S where substring sub is found, such that sub is contained within S[start:end].

Optional arguments start and end are interpreted as in slice notation.  
Return -1 on failure.

**Type:** method\_descriptor

```
In [121]: 1 print(f"1 + 1 = {1 + 1}")
```

```
1 + 1 = 2
```

```
In [122]: 1 bid = 1.5760
          2 ask = 1.5763
          3 print("bid: {}, ask: {}, spread: {}".format(bid, ask, ask - bid))
```

```
bid: 1.576, ask: 1.5763, spread: 0.000299999999999996696
```

```
In [123]: 1 print("bid: {:.4f}, ask: {:.4f}, spread: {:.4f}".format(bid, ask, ask - bid))
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [124]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {ask - bid:.4f}")
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [125]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {(ask - bid):.4f}")
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [126]: 1 string1 = "β, β, #"
          2 string1 = "Π, ú, θ, ω, ν"
```

```
3 chars = string1.split(", ")
4 print(chars)
```

```
['Π', 'ó', 'θ', 'ω', 'ν']
```

```
In [127]: 1 list1 = []
          2 for c in chars:
          3     list1.append(hex(ord(c)))
          4 print(list1)
```

```
['0x3a0', '0x3cd', '0x3b8', '0x3c9', '0x3bd']
```

```
In [128]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.upper())
          4 print(list1)
```

```
['Π', 'Υ', 'Θ', 'Ω', 'Ν']
```

```
In [129]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.lower())
          4 print(list1)
```

```
['π', 'ó', 'θ', 'ω', 'ν']
```

```
In [130]: 1 a = 42
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 42 is even
```

```
In [131]: 1 a = 41
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 41 is not even
```

```
In [132]: 1 a = 42
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 42 is even
```

```
In [133]: 1 a = 41
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 41 is not even
```

```
In [134]: 1 a = 0.3
          2 b = 0.1
          3 print(f"{a} / {b} = {a / b:.4f}")
```

```
0.3 / 0.1 = 3.0000
```

```
In [135]: 1 a = 42
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 42 is even

```
In [136]: 1 a = 421
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 421 is not even

```
In [137]: 1 print(range(10))
```

range(0, 10)

```
In [138]: 1 print(type(range(10)))
```

<class 'range'>

```
In [139]: 1 for i in range(3):
          2     print(i)
          3     print(i)
```

0  
1  
2  
2

```
In [140]: 1 for i in "abc":
          2     pass
          3     print(i)
```

c

```
In [141]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for r in range(len(raggedArray)):
          3     for c in range(len(raggedArray[r])):
          4         print(f"[{r}, {c}] = {raggedArray[r][c]}")
          5     print("-" * 5)
```

[0, 0] = 1  
[0, 1] = 2  
-----  
[1, 0] = 3  
[1, 1] = 4  
[1, 2] = 5  
[1, 3] = 6  
-----  
[2, 0] = 7  
[2, 1] = 8  
[2, 2] = 9  
-----  
[3, 0] = 10  
-----

```
In [142]: 1 identityMatrix = []
2 size = 10
3 for r in range(size):
4     row = []
5     for c in range(size):
6         if r == c:
7             row.append(1)
8         else:
9             row.append(0)
10    identityMatrix.append(row)
11    identityMatrix
```

```
Out[142]: [[1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
[0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
[0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
[0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
[0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
[0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
[0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 1, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 1]]
```

```
In [143]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
2 for i, r in enumerate(raggedArray):
3     for j, e in enumerate(raggedArray[i]): # e for element, cuz it's not a
4         column technically
5         print(f"[{i}, {j}] = {raggedArray[i][j]}")
6         print("-" * 8)
```

```
[0, 0] = 1
[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----
```

```
In [144]: 1 print(f"{0.1:.4f}")
```

```
0.1000
```

```
In [145]: 1 print("{:.4f}".format(0.1))
```

```
0.1000
```

```
In [146]: 1 total, count = (0,) * 2
2 data = [10.5, 11.2, 9.8, None, 11.5, None]
3 for e in data:
```

```
4         if e:
5             total += e
6             count += 1
7     average = total / count
8     print(average)
```

10.75

```
In [147]: 1 for i, e in enumerate(data):
2         if not e:
3             data[i] = average
4         print(data)
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [148]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2     count = sum(1 for e in data if e)
3     total = sum(e for e in data if e)
4     average = total / count
5     print(sum, average)
```

<built-in function sum> 10.75

```
In [149]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2     average = statistics.fmean(e for e in data if e)
3     data = [average if e is None else e for e in data]
4     print(data)
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [150]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2     df = pandas.DataFrame(data)
3     df.fillna(df.mean(), inplace = True)
4     print(df[0].tolist())
```

[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]

```
In [151]: 1 print(len(str((1, 2))))
```

6

```
In [152]: 1 print(str((1, 2)))
```

(1, 2)

```
In [153]: 1 m, n = 4, 5
2     for i in range(1, m):
3         for j in range(1, n):
4             print(f"{i} x {j} = {i * j}")
5     print("-" * 8)
```

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
-----
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
-----
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
-----
```

In [154]:

```
1
2
3 data = ("2021-01-01", 10, 20], ["2021-01-02", 20, 18], ["2021-01-03", -10,
4 10], ["2021-01-04", 100, 102], ["2021-01-05", 20, 45])
5 largest = 0
6 for e in data:
7     distance = abs(e[2] - e[1])
8     e.append(distance)
9     if distance > largest:
10         dateOfLargest = e[0]
11         largest = distance
12 print(data)
13 print(dateOfLargest)
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```

```
In [156]: 1 tuple1 = ([0],)
          2 print(type(tuple1))
          3 print(tuple1)
          4 print(id(tuple1))
          5 print(id(tuple1[0]))
          6 tuple1[0][0] += 1
          7 print(tuple1)
          8 print(id(tuple1))
          9 print(id(tuple1[0]))
```

```
<class 'tuple'>
([0],)
128573173295168
128573172611328
([1],)
128573173295168
128573172611328
```

```
In [157]: 1 print(hash(True))
          2 a = True
          3 print(hash(a))
          4 b = True
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
1
1
1
1
```

```
In [158]: 1 print(hash(257))
          2 a = 257
          3 print(hash(a))
          4 b = 257
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
257
257
257
257
```

```
In [159]: 1 print(hash(2570))
          2 a = 2570
          3 print(hash(a))
          4 b = 2570
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
2570
2570
2570
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```

```
In [160]: 1 print(hash(25))
          2 a = 25
          3 print(hash(a))
          4 b = 25
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
25
25
25
25
```

```
In [161]: 1 print(hash(2.5))
          2 a = 2.5
          3 print(hash(a))
          4 b = 2.5
          5 print(hash(b))
          6 print(hex(hash(b)))
          7 a = b
          8 print(len(hex(hash(a))))
          9 print(hash(b))
```

```
1152921504606846978
1152921504606846978
1152921504606846978
0x10000000000000002
18
1152921504606846978
```

```
In [162]: 1 for i in range(5):
          2     for j in range(4):
          3         pass
          4 print(i, j)
```

```
4 3
```

```
In [163]: 1 s1 = """
          2 "'And' and 'or' are the basic operations of logic. Together with 'no' (the
          3 logical operation
          4 of negation) they are a complete set of basic logical operations – all other
          5 logical
          6 operations, no matter how complex, can be obtained by suitable combinations of
          7 these."
          8 — John von Neumann, The Computer and the Brain
          9 """
          10
          11 s2 = """
          12 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
          13 incididunt ut
```

```

7 labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco laboris
8 nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit
9 esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat
1 non proident, sunt
0 in culpa qui officia deserunt mollit anim id est laborum.
1 """
1
1 charCounts = {}
2 for c in s1:
1     if (ord(c) >= ord("A") and ord(c) <= ord("Z")) or (ord(c) >= ord("a")
3 and ord(c) <= ord("z")):
1         charCounts[c] = charCounts.get(c, 0) + 1
4 print(charCounts)
1
5
1
6
1
7
1
8
1
9
2
0

```

```

{'A': 1, 'n': 20, 'd': 4, 'a': 23, 'o': 31, 'r': 12, 'e': 27, 't': 21, 'h': 11, 'b': 7, 's': 9, 'i':
17, 'c': 10, 'p': 7, 'f': 4, 'l': 12, 'g': 6, 'T': 2, 'w': 2, 'y': 2, 'm': 6, 'x': 1, 'u': 3, 'J':
1, 'v': 1, 'N': 1, 'C': 1, 'B': 1}

```

```

In [164]: 1 d1 = {"a": 10, "b": 20, "c": 30}
2 d2 = {"d": 100, "e": 200, "f": 300}
3 d3 = {"f": 30, "g": 40}
4 keysList, valuesList = [], []
5 for d in d1, d2, d3:
6     for k, v in d.items():
7         keysList.append(k)
8         valuesList.append(v)
9 print(keysList)
10 print(valuesList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]

```

```

In [165]: 1 keysList, valuesList = [], []
2 keysList = list(d1.keys())
3 keysList.extend(list(d2.keys()))
4 keysList.extend(list(d3.keys()))
5 print(keysList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']

```

```
In [166]: 1 valuesList = list(d1.values())
          2 valuesList.extend(list(d2.values()))
          3 valuesList.extend(list(d3.values()))
          4 print(valuesList)
```

```
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [167]: 1 print(list(d1.keys()) + list(d2.keys()) + list(d3.keys()))
          2 print(list(d1.values()) + list(d2.values()) + list(d3.values()))
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [168]: 1 grades = {"John": [90, 95, 98], "Eric": [86, 84, 92], "Michael": [90, 89, 85]}
          2 exam = {"Eric": 99, "John": 100}
          3 for k, v in grades.items():
          4     if k in exam:
          5         v.insert(0, exam[k])
          6     else:
          7         v.insert(0, None)
          8 print(grades)
```

```
{'John': [100, 90, 95, 98], 'Eric': [99, 86, 84, 92], 'Michael': [None, 90, 89, 85]}
```

```
In [169]: 1 for k, v in grades.items():
          2     score = exam.get(k)
          3     grades[k] = [score] + grades[k]
          4 print(grades)
```

```
{'John': [100, 100, 90, 95, 98], 'Eric': [99, 99, 86, 84, 92], 'Michael': [None, None, 90, 89, 85]}
```

```
In [170]: 1 for k, v in grades.items():
          2     v.insert(0, exam.get(k))
          3 print(grades)
```

```
{'John': [100, 100, 100, 90, 95, 98], 'Eric': [99, 99, 99, 86, 84, 92], 'Michael': [None, None, None, 90, 89, 85]}
```

```
In [171]: 1 for k in grades:
          2     print(k)
```

```
John
Eric
Michael
```

```
In [172]: 1 l = ["AAPL", "AAPL", "Aapl", "aapl", "MSFT"]
          2 print(list(set(l)))
```

```
['AAPL', 'Aapl', 'MSFT', 'aapl']
```

```
In [173]: 1 l = [e.casefold() for e in l]
          2 print(list(set(l)))
```

```
['aapl', 'msft']
```

```
In [174]: 1 se = {e.casefold() for e in l}
          2 print(se)
```

```
{'aapl', 'msft'}
```

```
In [175]: 1 data = {"d1": {"a": 1, "b": 2, "c": 3}, "d2": {"b": 20, "c": 30, "d": 40},
          2 "d3": {"d": 100, "x": 200}}
          3 uniqueKeys = {k for v in (data["d1"], data["d2"], data["d3"]) for k in v}
          4 print(uniqueKeys)
```

```
{'d', 'x', 'c', 'b', 'a'}
```

```
In [176]: 1 try:
          2     s1 = {[0, 1]}
          3     print(s1)
          4 except TypeError as ex:
          5     print(type(ex)., ":", " ", *ex.args, ".", sep = " ")
```

```
TypeError: cannot use 'list' as a set element (unhashable type: 'list').
```

```
In [177]: 1 m = [0] * 3
          2 print(m)
          3 print(m[0] is m[1])
```

```
[0, 0, 0]
True
```

```
In [178]: 1 m = [[0] * 3 for _ in range(3)]
          2 print(m)
          3 print(m[0] is m[1])
          4 print(m[0][0] is m[0][1])
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
False
True
```

```
In [179]: 1 m = [[10000 for i in range(3)] for _ in range(4)]
          2 print(m)
```

```
[[10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000]]
```

```
In [180]: 1 pprint.pprint(m, width = 1)
```

```
[[10000,
  10000,
  10000],
 [10000,
  10000,
  10000],
 [10000,
  10000,
  10000],
 [10000,
  10000,
  10000]]
```

```
In [181]: 1 print(m[0] is m[1])
          2 print(m[0][0] is m[0][1])
          3 print(hash(m[0][0]), hash(m[0][1]))
```

```
False
True
10000 10000
```

```
In [182]: 1 print(id(m[0][0]), id(m[0][1]))
```

```
128573172653872 128573172653872
```

```
In [183]: 1 m[0][0] = 1001
          2 pprint.pprint(m)
```

```
[[1001, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000]]
```

```
In [184]: 1 print(id(m[0][0]), id(m[0][1]))
          2 print(m[0][0] is m[0][1])
```

```
128573172653712 128573172653872
False
```

```
In [185]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = []
4 for d in data:
5     ranges.append(d["high"] - d["low"])
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [186]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = [d["high"] - d["low"] for d in data]
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [187]: 1 bogusComposites = {i for i in range(1, 101) for j in range(2, 10) if i % j ==
          2 0}
          3 allNumbers = {i for i in range(11, 101)} | {1}
          print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

```
In [188]: 1 allNumbers = set(range(1, 101))
          2 print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [189]:

```
data = [{"symbol": "ABCD", "name": "ABCD Company", "ranking": 2, "risk": 0.2},
{"symbol": "BCDE", "name": "BCDE Company", "ranking": 5, "risk": 0.2},
{"symbol": "CDEF", "name": "CDEF Company", "ranking": 8, "risk": 0.5},
{"symbol": "DEFG", "name": "DEFG Company", "ranking": 7, "risk": 0.8},
{"symbol": "EFGH", "name": "EFGH Company", "ranking": 9, "risk": 0.6},
{"symbol": "FGHI", "name": "FGHI Company", "ranking": 10, "risk": 0.1},
{"symbol": "GHIJ", "name": "GHIJ Company", "ranking": 3, "risk": 0.6},
{"symbol": "HIJK", "name": "HIJK Company", "ranking": 5, "risk": 0.5},
{"symbol": "IJKL", "name": "IJKL Company", "ranking": 5, "risk": 0.7},
{"symbol": "JKLM", "name": "JKLM Company", "ranking": 6, "risk": 0.9},
{"symbol": "KLMN", "name": "KLMN Company", "ranking": 6, "risk": 0.4},
{"symbol": "LMNO", "name": "LMNO Company", "ranking": 8, "risk": 0.4},
{"symbol": "MNOP", "name": "MNOP Company", "ranking": 8, "risk": 0.2},
{"symbol": "NOPQ", "name": "NOPQ Company", "ranking": 1, "risk": 0.5},
{"symbol": "OPQR", "name": "OPQR Company", "ranking": 9, "risk": 0.2},
{"symbol": "PQRS", "name": "PQRS Company", "ranking": 10, "risk": 0.9},
{"symbol": "QRST", "name": "QRST Company", "ranking": 3, "risk": 0.4},
{"symbol": "RSTU", "name": "RSTU Company", "ranking": 7, "risk": 0.3},
{"symbol": "STUV", "name": "STUV Company", "ranking": 8, "risk": 0.1},
{"symbol": "TUVW", "name": "TUVW Company", "ranking": 7, "risk": 0.9}]
filtered = [d for d in data if d["ranking"] > 5 and d["risk"] < 0.6]
print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1}]
```

In [190]:

```
1 for d in filtered:
2     d["weight"] = float(f"{d["ranking"] * d["risk"]:.2f}")
3 print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5, 'weight': 4.0}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1, 'weight': 1.0}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4, 'weight': 2.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4, 'weight': 3.2}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2, 'weight': 1.6}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2, 'weight': 1.8}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3, 'weight': 2.1}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1, 'weight': 0.8}]
```

In [191]:

```
1 weightedStocks = [dict(symbol = d["symbol"], weight = d["weight"]) for d in
2 filtered]
3 print(weightedStocks)
```

```
[{'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

In [192]:

```
1 weightedStocks = [dict(symbol = d["symbol"], weight = round(d["ranking"] *
2 d["risk"], 2)) for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
3 print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [193]: 1 weightedStocks = [{"symbol": d["symbol"], "weight": round(d["ranking"] *  
2 d["risk"], 2)} for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]  
print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [194]: 1 m = [[0] * 2] * 3  
2 print(m)  
3 for r in range(len(m)):  
4     for c in range(len(m[r])):  
5         print(id(m[r][c]))
```

```
[[0, 0], [0, 0], [0, 0]]  
11370064  
11370064  
11370064  
11370064  
11370064  
11370064
```

```
In [195]: 1 m = [[{0}] * 2] * 3  
2 print(m)  
3 for r in range(len(m)):  
4     for c in range(len(m[r])):  
5         print(id(m[r][c]))
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]  
128573172485248  
128573172485248  
128573172485248  
128573172485248  
128573172485248  
128573172485248
```

```
In [196]: 1 m[0][0] = 1  
2 for r in range(len(m)):  
3     for c in range(len(m[r])):  
4         print(id(m[r][c]))  
5 print(m)
```

```
11370096  
128573172485248  
11370096  
128573172485248  
11370096  
128573172485248  
[[1, {0}], [1, {0}], [1, {0}]]
```

```
In [197]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
          6 m[0][0] = 1
          7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
128573172485696
128573172482336
128573172482112
128573172486144
128573172487040
128573172485024
[[1, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [198]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
          6 m[0][0] = {1}
          7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
128573172485696
128573172483680
128573172487488
128573172486816
128573172486592
128573172488160
[[{1}, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [199]: 1 m = [[0 for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
11370064
11370064
11370064
11370064
11370064
11370064
```

```
In [200]: 1 m = [[10000 for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
          5 m[0][0] = 1
          6 print(m)
```

```
128573172653264
128573172653264
128573172653264
128573172653264
128573172653264
128573172653264
[[1, 10000], [10000, 10000], [10000, 10000]]
```

```
In [201]: 1 m = [{0} for _ in range(2)] for _ in range(3)
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
128573172483680
128573172487488
128573172485248
128573172486144
128573172488160
128573172486368
```

```
In [202]: 1 # print(input("Text box displayed ?"))
```

```
In [203]: 1 a = 0
          2 b = complex(real = 3.0, imag = 4)
          3 print(repr(a))
          4 print(repr(b))
          5 print(b)
```

```
0
(3+4j)
(3+4j)
```

```
In [204]: 1 values = [4, 3, 1, 2]
          2 try:
          3     minimum = abs(values[0])
          4     for value in values[1:]:
          5         if abs(value) < minimum:
          6             minimum = abs(value)
          7 except IndexError:
          8     minimum = 0
          9     pass
         10 finally:
         11     print(f"Minimum is: {minimum}")
```

```
Minimum is: 1
```

```
In [205]: 1 values = []
          2
          3 try:
          4     minimum = abs(values[0])
          5     for value in values[1:]:
          6         if abs(value) < minimum:
          7             minimum = abs(value)
          8 except IndexError:
          9     minimum = 0
```

```
10     pass
11 finally:
12     print(f"Minimum is: {minimum}")
```

Minimum is: 0

```
In [206]: 1 try:
2         raise ValueError("Custom message")
3 except ValueError as ex:
4     print(type(ex).      + ": " + ex.args[0] + ".")
```

ValueError: Custom message.

```
In [ ]: 1 l = [4, 3, 2, 1, 0]
2 itr = iter(l)
3 try:
4     while True:
5         print(next(itr))
6 except StopIteration as ex:
7     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
4
3
2
1
0
StopIteration: .
```

```
In [208]: 1 try:
2         raise IndexError
3 except IndexError as ex:
4     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

IndexError: .

```
In [209]: 1 rangeObject = range(10)
2 rangeIterator = iter(rangeObject)
3 print(next(rangeIterator))
4 print(next(rangeIterator))
5 print(next(rangeIterator))
6 print(list(rangeIterator))
```

```
0
1
2
[3, 4, 5, 6, 7, 8, 9]
```

```
In [210]: 1 rangeObj = range(5)
2 print(list(rangeObj))
3 print(list(rangeObj))
```

```
[0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
```

```
In [211]: 1 enumIterator = enumerate(string.ascii_lowercase)
2 print(enumIterator)
```

```

3 l = list(enumIterator)
4 print(l)
5 l = list(enumIterator)
6 print(dir(type(iter([0]))))
7 print(l)
8 try:
9     print(next(enumIterator))
10 except StopIteration as ex:
11     print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

```

<enumerate object at 0x74efc6458680>
[(0, 'a'), (1, 'b'), (2, 'c'), (3, 'd'), (4, 'e'), (5, 'f'), (6, 'g'), (7, 'h'), (8, 'i'), (9, 'j'),
(10, 'k'), (11, 'l'), (12, 'm'), (13, 'n'), (14, 'o'), (15, 'p'), (16, 'q'), (17, 'r'), (18, 's'),
(19, 't'), (20, 'u'), (21, 'v'), (22, 'w'), (23, 'x'), (24, 'y'), (25, 'z')]
['_class_', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__',
'__getattr__', '__getstate__', '__gt__', '__hash__', '__init__', '__init_subclass__',
'__iter__', '__le__', '__length_hint__', '__lt__', '__ne__', '__new__', '__next__', '__reduce__',
'__reduce_ex__', '__repr__', '__setattr__', '__setstate__', '__sizeof__', '__str__',
'__subclasshook__']
[]
StopIteration: .

```

```

In [212]: 1 print(type(enumIterator))
          2 print(type(iter(enumIterator)))
          3 print(issubclass(type(iter([0])), type(enumIterator)))

```

```

<class 'enumerate'>
<class 'enumerate'>
False

```

```

In [213]: 1 print(iter(enumIterator) is enumIterator)

True

```

```

In [214]: 1 print(type(iter({0, 1, 2})))

<class 'set_iterator'>

```

```

In [215]: 1 print(type(iter({0: "0", 1: "1", "2": 2})))

<class 'dict_keyiterator'>

```

```

In [ ]: 1 l = [4, 3, 2, 1, 0]
        2 itr = iter(l)
        3 for i in itr:
        4     print(i)

```

```

4
3
2
1
0

```

```

In [217]: 1 try:
          2     print(next(l))
          3 except TypeError as ex:
          4     print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

TypeError: 'list' object is not an iterator.

```
In [218]: 1 start = time.perf_counter()
          2 squaresGenerator = (i ** 10 for i in range(1_000_000))
          3 end = time.perf_counter()
          4 print(f"Elapsed: {end - start}")
```

Elapsed: 0.00016390199999705146

```
In [219]: 1 start = time.perf_counter()
          2 squaresListComp = [i ** 10 for i in range(1_000_000)]
          3 end = time.perf_counter()
          4 print(f"Elapsed: {end - start}")
```

Elapsed: 0.23438802800001213

```
In [220]: 1 print((7.16e-5 / 1.824e-1) ** -1)
          2 print(105 / 9)
          3 print(10.5e1 / 9e0)
          4 print(99 / 1)
          5 print(9.9e1 / 1e0) # 1 order of magnitude difference means a multiple of
                                between 10 and 100
```

2547.486033519553  
11.666666666666666  
11.666666666666666  
99.0  
99.0

```
In [221]: 1 print(999_999 ** 10)
```

99999000000449998800002099997480002099998800000449999900000001

```
In [222]: 1 print(squaresGenerator is iter(squaresGenerator))
```

True

```
In [223]: 1 l = [10, "abc", 3.14, True]
          2 for i, e in enumerate(l):
          3     print(f"{i}: {e}")
```

0: 10  
1: abc  
2: 3.14  
3: True

```
In [224]: 1 def f(a, *b, x, **y):
          2     print("Positional/Named:", a)
          3     print("Variable positional:", b)
          4     print("Named only:", x)
          5     print("Variable named:", y)
          6
          7
          8 f(0, 1, 2, x = 3, y = 4, z = 5)
```

Positional/Named: 0  
Variable positional: (1, 2)  
Named only: 3  
Variable named: {'y': 4, 'z': 5}

```
In [225]: 1 # f(10, 2, x = 10, 20)
```

```
In [226]: 1 f(10, 2, x = 10, y = 20)
```

Positional/Named: 10  
Variable positional: (2,)  
Named only: 10  
Variable named: {'y': 20}

```
In [227]: 1 f(10, z = 2, x = 10, y = 20)
```

Positional/Named: 10  
Variable positional: ()  
Named only: 10  
Variable named: {'z': 2, 'y': 20}

```
In [228]: 1 f(10, x = 2, y = 10, z = 20)
```

Positional/Named: 10  
Variable positional: ()  
Named only: 2  
Variable named: {'y': 10, 'z': 20}

```
In [229]: 1 # f(a = 10, 2, x = 10, y = 20)
```

```
In [230]: 1 f(x = 10, y = 20, z = 30, a = 10)
```

Positional/Named: 10  
Variable positional: ()  
Named only: 10  
Variable named: {'y': 20, 'z': 30}

```
In [231]: 1 try:
2         f(10, a = 2, x = 10, y = 20)
3     except TypeError as ex:
4         print(type(ex).      )
```

TypeError

```
In [232]: 1 def f1(a, *, x, **y):
2         print(f"a: {a}, x: {x}, y: {y}")
```

```
In [233]: 1 try:
2         f1(10, a = 20, x = 2, y = 10)
3     except TypeError as ex:
4         print(type(ex).      )
```

TypeError

```
In [234]: 1 f1(a = 1, x = 2)
```

```
a: 1, x: 2, y: {}
```

```
In [235]: 1 print(max("cba"))
```

```
c
```

```
In [236]: 1 def f2(a, b):  
2         print(f"a: {a}; b: {b}")  
3  
4  
5 f2(2, 3)
```

```
a: 2; b: 3
```

```
In [237]: 1 f2(2, b = 3)
```

```
a: 2; b: 3
```

```
In [238]: 1 # f2(a = 2, 3)
```

```
In [239]: 1 f2(b = 1, a = 0)
```

```
a: 0; b: 1
```

```
In [240]: 1 # def f3(a = 0, b):  
2         # print(f"a: {a}; b: {b}")
```

```
In [241]: 1 def f3(a, b = 0):  
2         print(f"a: {a}; b: {b}")  
3  
4  
5 f3(-1, 1)
```

```
a: -1; b: 1
```

```
In [242]: 1 # f3(a = -1, 1)
```

```
In [243]: 1 f3(a = -1, b = 1)
```

```
a: -1; b: 1
```

```
In [244]: 1 def f4(a, b = 1, *, x):  
2         print(a, b, x)  
3  
4  
5 f4(0, 2, x = 3)
```

```
0 2 3
```

```
In [245]: 1 def f5(a, b = 1, *, x = 2, y):  
2         print(a, b, x, y)  
3  
4  
5 f5(0, 2, x = 3, y = 4)
```

0 2 3 4

```
In [246]: 1 def f6(**x):
          2     print(x)
          3
          4
          5 f6(a = 1, b = 2, c = 3)
```

{'a': 1, 'b': 2, 'c': 3}

```
In [247]: 1 # def f7(*, **x):
          2 #     print(x)
          3 #
          4 #
          5 # f7(a = 1, b = 2)
```

```
In [248]: 1 def f8(a, *, b):
          2     print(f"a: {a}; b: {b}")
          3
          4
          5 f8(b = 1, a = 0)
```

a: 0; b: 1

```
In [249]: 1 def toJSON(a, *, b, **c):
          2     print(a, b, c)
          3
          4
          5 toJSON(0, b = 3, c1 = 0, c2 = 1)
```

0 3 {'c1': 0, 'c2': 1}

```
In [250]: 1 def avg(a, *b):
          2     return (sum(b) + a) / (len(b) + 1)
          3
          4
          5 print(avg(0, 1, 2, 3, 4))
          6 print(avg(0))
          7 print(avg(0, 1))
```

2.0  
0.0  
0.5

```
In [251]: 1 def separator(b = 10, *, a = "-"):
          2     return a * b
          3
          4
          5 print(separator())
          6 print(separator(a = "A"))
          7 print(separator(5))
          8 print(separator(a = "B", b = 5))
          9 print(separator(5, a = "B"))
```

```
-----  
AAAAAAAAA  
-----  
BBBBBB  
BBBBBB
```

```
In [252]: 1 print(sep = "|", *("AB", "CD"))
```

```
AB|CD
```

```
In [253]: 1 # def f(a, *b, c):  
2         #     print(a, b, c)  
3         #  
4         #  
5         # f(c = 2, 0, 1, 2)
```

```
In [254]: 1 # def f(*a, b = 0):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0, 2)
```

```
In [255]: 1 try:  
2         f(b = 1, *(0, 2))  
3 except TypeError as ex:  
4         print(type(ex).      )
```

```
TypeError
```

```
In [256]: 1 # def f(a, *, b):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [257]: 1 # def f(a, b):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [258]: 1 # def f(a, b = 2):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [259]: 1 def f(a, b = 2):  
2         print(a, b)  
3
```

```
4
5 f(b = 1, *(0,))
```

```
0 1
```

```
In [260]: 1 anonymous = lambda iterable: len(set(iterable))
2 print(anonymous("abcd"))
3 print(anonymous("abab"))
4 try:
5     print(anonymous())
6 except TypeError as ex:
7     print(type(ex).      )
8 print(anonymous(""))
9 print(anonymous([0, 1, 2]))
10 print(anonymous([0]))
11 print(anonymous([]))
12 print(anonymous([1, 1, 2, 2, 3, 4]))
```

```
4
2
TypeError
0
3
1
0
4
```

```
In [261]: 1 s = "This is the first sentence. This is the scecond sentence. This is not the
2 fourth sentence, it is the third sentence."
3
4
5 def f(s = ""):
6     return dict(collections.Counter(s))
7
8
9 print(f(s))
print(f(""))
```

```
{'T': 3, 'h': 9, 'i': 10, 's': 13, ' ': 20, 't': 13, 'e': 17, 'f': 2, 'r': 3, 'n': 10, 'c': 6, 'l':
3, 'o': 3, 'd': 2, 'u': 1, ',': 1}
{}
```

```
In [262]: 1 s = "This is the first sentence. This is the scecond sentence. This is not the
2 fourth sentence, it is the third sentence."
3
4 def f(s = ""):
5     words = re.split("\\.| |", s)
6     for word in words:
7         if word == "":
8             words.remove(word)
9     frequencies = dict.fromkeys(set(words), 0)
10    for word in words:
11        frequencies[word] += 1
12    print(frequencies)
```

```
7
8 f(s)
9
10
11
12
13
14
15
```

```
{'is': 4, 'the': 4, 'fourth': 1, 'third': 1, 'it': 1, 'not': 1, 'first': 1, 'second': 1, 'sentence': 4, 'This': 3}
```

```
In [263]: 1 def f(s = ""):
2         s = s.replace(".", "")
3         s = s.replace(",", "")
4         words = s.split()
5         return dict(collections.Counter(words))
6
7
8 print(f(s))
```

```
{'This': 3, 'is': 4, 'the': 4, 'first': 1, 'sentence': 4, 'second': 1, 'not': 1, 'fourth': 1, 'it': 1, 'third': 1}
```

```
In [264]: 1 print(round(1.812))
```

```
2
```

```
In [265]: 1 n, d = (0.3).as_integer_ratio()
2         print(n, d)
3         print(1 / 3)
```

```
5404319552844595 18014398509481984
0.3333333333333333
```

```
In [266]: 1 print(round(0.325, 2))
```

```
0.33
```

```
In [267]: 1 print(f"{0.325:.40f}")
```

```
0.3250000000000000111022302462515654042363
```

```
In [268]: 1 def toJSON(a, *, b, **c):
2         print(a, b, c)
3
```

```

4
5 try:
6     toJSONN(0, 1, 2, b = 3, c1 = 4, c2 = 5)
7 except TypeError as ex:
8     print(type(ex).

```

TypeError

```

In [269]: 1 def toJSONN(a, *, b, **c):
2         print(a, b, c)
3
4
5 toJSONN(0, b = 3, c1 = 4, c2 = 5)

```

0 3 {'c1': 4, 'c2': 5}

```

In [270]: 1 # def toJSONN(a, *):
2         #     print(a)
3         #
4         #
5         # toJSONN(0, 1, 2)

```

```

In [271]: 1 def toJSONN(a, *args):
2         print(a, args)
3
4
5 toJSONN(0, 1, 2)

```

0 (1, 2)

```

In [272]: 1 l = ["AB CD", "EF CC", "GH BC", "IJ AC", "KL CA", "MN JQ", "OP BA"]
2         print(sorted(l, key = lambda e: (e[3], e[4])))

```

['IJ AC', 'OP BA', 'GH BC', 'KL CA', 'EF CC', 'AB CD', 'MN JQ']

```

In [273]: 1 # print(sorted(l, key = lambda e: e[len(e) - i] for i in range(len(e))))
2         print(sorted(l, key = lambda e: e[len(e) - 1]))

```

['KL CA', 'OP BA', 'EF CC', 'GH BC', 'IJ AC', 'AB CD', 'MN JQ']

```

In [274]: 1 a = (1 for _ in range(5))
2         print(a)
3         print(lambda _: "ABCD")
4         print(sorted)
5         print(range)

```

<generator object <genexpr> at 0x74efc64a24d0>  
 <function <lambda> at 0x74efc644b740>  
 <built-in function sorted>  
 <class 'range'>

```

In [275]: 1 def f(*a, b):
2         print(a, b)
3

```

```
4
5 f(b = 0, *(1, 2))
```

```
(1, 2) 0
```

```
In [276]: 1 def f(a, *b, c):
          2     print(a, b, c)
          3
          4
          5 f(0, c = 1, *(2, 3))
```

```
0 (2, 3) 1
```

```
In [277]: 1 print(statistics.fmean)
```

```
<function fmean at 0x74effaaa6a30>
```

```
In [278]: 1 print(dict([("a", 97), ("b", 98), ("c", 99)]))
```

```
{'a': 97, 'b': 98, 'c': 99}
```

```
In [279]: 1 widgets = [f"w{i}" for i in range(1, 21)]
          2 skus = [f"sku{i}" for i in range(1, len(widgets) + 1)]
          3 print(widgets)
          4 print(skus)
```

```
['w1', 'w2', 'w3', 'w4', 'w5', 'w6', 'w7', 'w8', 'w9', 'w10', 'w11', 'w12', 'w13', 'w14', 'w15',
'w16', 'w17', 'w18', 'w19', 'w20']
['sku1', 'sku2', 'sku3', 'sku4', 'sku5', 'sku6', 'sku7', 'sku8', 'sku9', 'sku10', 'sku11', 'sku12',
'sku13', 'sku14', 'sku15', 'sku16', 'sku17', 'sku18', 'sku19', 'sku20']
```

```
In [280]: 1 print({w: s for w, s in zip(widgets, skus)})
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [281]: 1 print(dict(zip(widgets, skus)))
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [282]: 1 suits = "shdc" # Spades, Hearts, Diamonds, Clubs
          2 ranks = list("23456789") + ["10", "J", "Q", "K", "A"]
          3 print(suits)
          4 print(ranks)
```

```
shdc
['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']
```

```
In [283]: 1 print([(rank, suit) for rank in ranks] for suit in suits))
```

```
[('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's')], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

In [284]:

1

```
pprint.pprint([(rank, suit) for rank in ranks] for suit in suits))
```

```
[('2', 's'),
 ('3', 's'),
 ('4', 's'),
 ('5', 's'),
 ('6', 's'),
 ('7', 's'),
 ('8', 's'),
 ('9', 's'),
 ('10', 's'),
 ('J', 's'),
 ('Q', 's'),
 ('K', 's'),
 ('A', 's')],
 [('2', 'h'),
 ('3', 'h'),
 ('4', 'h'),
 ('5', 'h'),
 ('6', 'h'),
 ('7', 'h'),
 ('8', 'h'),
 ('9', 'h'),
 ('10', 'h'),
 ('J', 'h'),
 ('Q', 'h'),
 ('K', 'h'),
 ('A', 'h')],
 [('2', 'd'),
 ('3', 'd'),
 ('4', 'd'),
 ('5', 'd'),
 ('6', 'd'),
 ('7', 'd'),
 ('8', 'd'),
 ('9', 'd'),
 ('10', 'd'),
 ('J', 'd'),
 ('Q', 'd'),
 ('K', 'd'),
 ('A', 'd')],
 [('2', 'c'),
 ('3', 'c'),
 ('4', 'c'),
 ('5', 'c'),
 ('6', 'c'),
 ('7', 'c'),
 ('8', 'c'),
 ('9', 'c'),
 ('10', 'c'),
 ('J', 'c'),
 ('Q', 'c'),
 ('K', 'c'),
 ('A', 'c')]]
```

```
In [285]: 1 print([list(zip(ranks, s * 13)) for s in suits])
```

```
[(('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's'))], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

In [286]:

```
1 def f(list1, *, reverse = False):
2     if not reverse:
3         return sorted(list1)
4     else:
5         return sorted(list1, reverse = True)
6
7
8 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
9 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
10 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
11 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [287]:

```
1 def f(list1, *, reverse = False):
2     return sorted(list1, reverse = reverse)
3
4
5 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
6 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
7 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
8 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [288]:

```
1 print(hex(ord(chr(0xE3).upper())))
```

0xc3

In [289]:

```
1 print("abc" in "defabcghi")
```

True

In [290]:

```
1 print("abc" in "abbde")
```

False

In [291]:

```
1 def solve(a, b, c):
2     solution = ((-1 * b) + math.sqrt((b * b) - (4 * a * c))) / (2 * a)
3     return round(solution, 2)
4
5
```

```
6 print(solve(3, 4, -5))
7 print(solve(1, -5, -8))
```

```
0.79
6.27
```

```
In [292]: 1 def encrypt(s):
          2     return "".join(chr(ord(c) + 10) for c in s)
          3
          4
          5 print(encrypt("ABCDEFGHIJKLMNOPQRSTUVWXYZ"))
```

```
KL MN OP QR ST UV WX YZ [\]^_`abcd
```

```
In [293]: 1 def decrypt(s):
          2     return "".join(chr(ord(c) - 10) for c in s)
          3
          4
          5 print(decrypt("KL MN OP"))
          6 print(decrypt("S}kkm* Xo\x81~yx"))
```

```
ABCDEF
Isaac Newton
```

```
In [294]: 1 print(10 % 3)
```

```
1
```

```
In [295]: 1 a = array.array("l", [20])
          2 print(a)
```

```
array('l', [20])
```

```
In [296]: 1 for i in a:
          2     print(i)
```

```
20
```

```
In [297]: 1 print(sys.getsizeof(a))
```

```
88
```

```
In [298]: 1 try:
          2     print(sys.getallocatedblocks(a))
          3 except TypeError as e:
          4     print(type(e).      )
```

```
TypeError
```

```
In [299]: 1 print(sys.getsizeof(1))
```

```
28
```

```
In [300]: 1 def f(s):
          2     print(s)
          3
```

```

4
5 try:
6     f(*"hello")
7 except TypeError as ex:
8     print(type(ex).      , ex.args)

```

TypeError ('f()' takes 1 positional argument but 5 were given',)

```

In [301]: 1 def func(s):
2           print(s)
3
4
5 def apply(func, *str):
6     result = func(*str)
7     return result
8
9
10 apply(func, "john")

```

john

```

In [302]: 1 try:
2           apply(func, "john", "smith")
3 except TypeError as ex:
4     print(type(ex).      , *ex.args)

```

TypeError func() takes 1 positional argument but 2 were given

```

In [303]: 1 def encloseGetPower(b):
2           print("hex(id(b)):", hex(id(b)))
3
4           def getPower(e):
5               print("hex(id(e)):", hex(id(e)))
6
7               return b ** e
8
9           print("getPower:", getPower)
10
11          return getPower
12
13
14 getBinaryPower = encloseGetPower(2)
15 print(getBinaryPower(31))
16 getTertiaryPower = encloseGetPower(3)
17 print(getTertiaryPower(4))

```

```

hex(id(b)): 0xad7e90
getPower: <function encloseGetPower.<locals>.getPower at 0x74efc644bd70>
hex(id(e)): 0xad8230
2147483648
hex(id(b)): 0xad7eb0
getPower: <function encloseGetPower.<locals>.getPower at 0x74efc644be20>
hex(id(e)): 0xad7ed0
81

```

```
In [304]: 1 print("getBinaryPower:", getBinaryPower)
          2 print("encloseGetPower:", encloseGetPower)
          3 try:
          4     print("getPower:", getPower)
          5 except NameError as ex:
          6     print(type(ex).      , ex.args)
```

```
getBinaryPower: <function encloseGetPower.<locals>.getPower at 0x74efc644bd70>
encloseGetPower: <function encloseGetPower at 0x74efc644b5e0>
NameError ("name 'getPower' is not defined",)
```

```
In [305]: 1 print("getBinaryPower:", getBinaryPower.      )
          2 print("encloseGetPower:", encloseGetPower.      )
```

```
getBinaryPower: (<cell at 0x74efc24f1510: int object at 0xad7e90>,)
encloseGetPower: None
```

```
In [306]: 1 def outer(a, b):
          2     print(hex(id(a)))
          3     sumTotal = a + b
          4
          5     def inner():
          6         prod = a * b
          7         print(a, b, sumTotal, prod)
          8         return "You just called a closure!"
          9
         10     return inner
         11
         12
         13 f = outer(2, 3)
```

```
0xad7e90
```

```
In [307]: 1 f = outer(2, 3)
```

```
0xad7e90
```

```
In [308]: 1 print(f)
```

```
<function outer.<locals>.inner at 0x74efc644bed0>
```

```
In [309]: 1 print(f.      )
```

```
(<cell at 0x74efc641f430: int object at 0xad7e90>, <cell at 0x74effaaf9db0: int object at 0xad7eb0>,
<cell at 0x74effb46b430: int object at 0xad7ef0>)
```

```
In [310]: 1 f()
```

```
2 3 5 6
```

```
Out[310]: 'You just called a closure!'
```

```
In [311]: 1 def execute(func):
          2
          3     def inner(a, b):
```

```

4         return func(a, b)
5
6     return inner
7
8
9 def add(a, b):
10     return a + b
11
12
13 add_executor = execute(add)
14 print(add_executor.

```

(<cell at 0x74efc24f0a30: function object at 0x74efc24f8040>,)

In [312]: 1 print(hex(id(add)))

0x74efc24f8040

In [313]: 1 try:  
2 print(hex(add))  
3 except TypeError as ex:  
4 print(type(ex).)

TypeError

In [314]: 1 add\_executor(2, 3)

Out[314]: 5

In [315]: 1 def f():  
2 pass  
3  
4  
5 print(f.)

None

In [316]: 1 def f():  
2  
3 def g():  
4 return  
5  
6 return g  
7  
8  
9 h = f()  
10 print(f.)  
11 print(h.)

None

None

In [317]: 1 a = 0  
2 try:

```
3     print(a.          )
4 except AttributeError as e:
5     print(type(e).    )
```

AttributeError

```
In [318]: 1 a = []
          2 try:
          3     print(a.          )
          4 except AttributeError as e:
          5     print(type(e).    )
```

AttributeError

```
In [319]: 1 def execute(func):
          2
          3     def inner(*posargs, **kwargs):
          4         return func(*posargs, **kwargs)
          5
          6     return inner
          7
          8
          9 def add(a, b, c):
          10     return a + b + c
          11
          12
          13 def say_hello(name, *, formal = True):
          14     return f"Pleased to meet you, {name}" if formal else f"Hi, {name}!"
          15
          16
          17 exec_add = execute(add)
          18 exec_greet = execute(say_hello)
```

```
In [320]: 1 exec_add(1, 2, 3)
```

Out[320]: 6

```
In [321]: 1 exec_greet("Michael", formal = False)
```

Out[321]: 'Hi, Michael!'

```
In [322]: 1 exec_greet("Michael", formal = True)
```

Out[322]: 'Pleased to meet you, Michael'

```
In [323]: 1 print((0.1).as_integer_ratio())
```

(3602879701896397, 36028797018963968)

```
In [324]: 1 print((0.125).as_integer_ratio())
```

(1, 8)

```
In [325]: 1 print((0.2).as_integer_ratio())
          2 print(f"{0.2:.40f}")
```

```
(3602879701896397, 18014398509481984)
0.2000000000000000111022302462515654042363
```

```
In [326]: 1 print(math.pi.as_integer_ratio())
          2 n, d = math.pi.as_integer_ratio()
          3 print(n / d)
```

```
(884279719003555, 281474976710656)
3.141592653589793
```

```
In [327]: 1 try:
          2     a1 = a1 + 1
          3     print(a1)
          4 except NameError as ex:
          5     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
NameError: name 'a1' is not defined.
```

```
In [328]: 1 def getDiagonalMatrix(val, rows, columns):
          2     diagonalMatrix = [[val if ce == r else 0 for ce in range(4)] for r in
          3     range(5)]
          4     return diagonalMatrix
```

```
In [329]: 1 def getDiagonalMatrix(rows, columns, *, val):
          2     diagonalMatrix = [[val if cl == row else 0 for cl in range(columns)]
          3     for row in range(rows)]
          4     return diagonalMatrix
```

```
In [330]: 1 def getFactorial(n):
          2     factorial = 1
          3     for i in range(1, n + 1):
          4         factorial *= i
          5     return factorial
```

```
In [331]: 1 result = getFactorial(5)
          2 print(result)
```

```
120
```

```
In [332]: 1 result = getDiagonalMatrix(3, 4, val = 5)
          2 pprint.pprint(result, width = 14)
```

```
[[5, 0, 0, 0],
 [0, 5, 0, 0],
 [0, 0, 5, 0]]
```

```
In [333]: 1 print(f"{timeit.timeit('getFactorial(2)', globals = globals()):.4g}")
```

```
0.1758
```

In [334]:

```
1 n = 0.000000000000123456789
2 n = 4.482882e-05
3 exp = 0
4 while n < 1:
5     exp -= 3
6     n *= 1000
7 prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f", "a"
8 print(f"{n}{prefixes[abs(exp)//3]}")
```

44.82882μ

In [335]:

```
1 n = 1000000000
2 exp = 0
3 while n >= 1000:
4     exp += 3
5     n /= 1000
6 prefixes = "", "k", "M", "G", "T", "P", "E"
7 print(f"{n}{prefixes[exp//3]}")
```

1.0G

In [7]:

```
def toEnggNotation(n, scientificNotation = True, nSignificantDigits = 4):
1     exp = 0
2     oldN = n
3     if n > 1:
4         while n >= 1000:
5             exp += 3
6             n /= 1000
7         prefixes = "", "k", "M", "G", "T", "P", "E"
8         return f"{n:.3f}{prefixes[exp//3]}\t{oldN}" if
scientificNotation else f"{n}{prefixes[exp//3]}"
9     elif n < 1:
10        while n < 1:
11            exp -= 3
12            n *= 1000
13        prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f",
14        "a"
15        return f"{n:.{nSignificantDigits}f}
16        {prefixes[abs(exp)//3]}\t{oldN}" if scientificNotation else f"{n}
17        {prefixes[abs(exp)//3]}"
18    else:
19        return 1
```

```
1 def myTimer(f, *pos, **keyword):
2     start = time.perf_counter()
3     print(f(*pos, **keyword))
4     end = time.perf_counter()
5     return end - start
6
7
```

```
8 print(toEnggNotation(myTimer(getFactorial, 1000)))
9 print(toEnggNotation(myTimer(getDiagonalMatrix, 3, 5, val = 7)))
```

[illegible]

```

1 def encloseMyTimer(f):
2
3     def myTimer(*pos, **keyword):
4         start = time.perf_counter()
5         result = f(*pos, **keyword)
6         end = time.perf_counter()
7         print(toEnggNotation(end - start, True, 4))
8         return result
9
10    return myTimer
11
12
13 timed = encloseMyTimer(getDiagonalMatrix)
14 print(timed(2, 4, val = -5))
15 timed = encloseMyTimer(getFactorial)
16 print(timed(11))
17 print(encloseMyTimer(getDiagonalMatrix)(2, 4, val = -5))
18 print(encloseMyTimer(getFactorial)(11))

```

```
5.2100µ 5.209999983435409e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
4.1080µ 4.108000041469495e-06
39916800
3.6170µ 3.6170000043966866e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
2.4240µ 2.4240000016106933e-06
39916800
```

```
In [341]: 1 # print(eval("2 * 3"))
          2 # print(eval("x = 2 and print(x)"))
```

```
In [342]: 1 # eval("x = 2")
          2 # print(x)
```

```
In [343]: 1 x = 2
          2 eval("x ** 2 - 1")
```

```
Out[343]: 3
```

```
In [344]: 1 def f(expr, x):
          2     print(eval(expr))
          3
          4
          5 x = 2
          6 f("x ** 2 - 1", x)
```

```
3
```

```
In [345]: 1 def f(x):
          2     return True
          3
          4
          5 try:
          6     print(list(map(f(False), range(0, 5))))
          7 except TypeError as ex:
          8     print(type(ex).      + ":", ex.args[0])
```

```
TypeError: 'bool' object is not callable
```

```
In [346]: 1 def getExtremum(expr, interval, granularity, minimax):
          2
          3     def f(expr):
          4
          5         def g(x):
          6             return eval(expr)
          7
          8         return g
          9
          10    evalExpr = f(expr)
          11    print("Range:", list(range(interval[0], interval[1] + granularity,
          12    granularity)))
          13    if minimax == "minimum":
```

```

7         return min(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
8     elif minimax == "maximum":
9         return max(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
10    else:
11        return "Invalid minimax"
12
13    expr = "x ** 2 - 1"
14    print(getExtremum(expr, (-10, 10), 2, "minimum"))
15    print(getExtremum(expr, (-10, 10), 2, "maximum"))
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96
97
98
99

```

```

Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
-1
Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
99

```

```

In [347]: 1 expr = "x ** 2 - 1"
          2 f = lambda x: eval(expr)
          3 print(f(3))

```

8

```

In [348]: 1 def getExtremum(expr, interval, resolution, minimax):
          2     def f(expr):
          3         def g(x):
          4             return eval(expr)
          5         return g
          6     evalExpr = f(expr)

```

```

    l = [interval[0] + (i * (interval[1] - interval[0]) / (resolution -
6    1)) for i in range(resolution)] # Cheated with the calc.
    return min(map(evalExpr, l)) if minimax == "minimum" else
7    max(map(evalExpr, l))

8
    expr = "x ** 2 - 1"
9    print(getExtremum(expr, (-10, 10), 5, "minimum"))
1    print(getExtremum(expr, (-10, 10), 5, "maximum"))
0    print(getExtremum(expr, (-10, 10), 6, "minimum"))
1    print(getExtremum(expr, (-10, 10), 6, "maximum"))
1    expr = "math.log10(x)"
1    print(getExtremum(expr, (1, 100), 20, "minimum"))
2    print(getExtremum(expr, (1, 100), 20, "maximum"))
1    expr = "math.sin(x)"
3    print(getExtremum(expr, (-10, 10), 20, "minimum"))
1    print(getExtremum(expr, (-10, 10), 20000, "maximum"))
4    expr = "x"
1    print(getExtremum(expr, (1, 6), 8, "minimum"))
5    print(getExtremum(expr, (1, 6), 8, "maximum"))

1
6
1
7
1
8
1
9
2
0
2
1
2
2
2
3
2
4
2
5
2
6
2
7
2
8

```

```
-1.0
99.0
3.0
99.0
0.0
2.0
-0.9999667804441291
0.999999996044897
1.0
6.0
```

In [349]:

```
def f(t):
    return t[0] ** t[1]

l = [(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3),
      (4, 4)]

start = time.perf_counter()
for _ in range(10_000):
    result = []
    for t in l:
        result.append(f(t))
end = time.perf_counter()
forLoopTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = [f(t) for t in l]
end = time.perf_counter()
listCompTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = list(map(f, l))
end = time.perf_counter()
mapTime = end - start
print(result)

print(toEnggNotation(forLoopTime), toEnggNotation(listCompTime),
      toEnggNotation(mapTime))
```

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0

```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
15.1299m      0.0151298600000001827  10.1019m   0.010101901999973961  10.2137m   0.010213654999972732
```

In [350]:

```
def power(x, y):
    return x ** y

def dist(pt1, pt2):
    return math.sqrt(sum(coord1 - coord2 for coord1, coord2 in zip(pt1,
pt2)))

def partial(f, operand2):

    def calcPartialFunction(operand1):
        return f(operand1, operand2)

    return calcPartialFunction

squares = partial(power, 2)
print(squares(3))
print(squares(4))
dist_from_origin = partial(dist, (0, 0))
print(dist_from_origin((1, 1)))
gcd_60 = partial(math.gcd, 60)
print(gcd_60(13))
log_2 = partial(math.log, 2)
print(log_2(8))
log_10 = partial(math.log, 10)
print(log_10(100))
```

```
5 log_16 = partial(math.log, 16)
6 print(log_16(65536))
```

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```

```
9
16
1.4142135623730951
1
3.0
2.0
4.0
```

In [351]:

```
1 def norm(x, y):
2     return math.sqrt(x ** 2 + y ** 2)
3
4
5 def find_index_min(seq):
6     minima = min(seq)
7     return seq.index(minima)
8
9
10 def logged(f):
11
12     def g(*args):
13         print("Logging...")
14         start = time.perf_counter()
15         print(f(*args))
```

```

16         end = time.perf_counter()
17         print(toEnggNotation(end - start))
18         print("... Finished")
19
20     return g
21
22
23 loggedNorm = logged(norm)
24 loggedNorm(3, 4)
25 loggedFindIndexMin = logged(find_index_min)
26 loggedFindIndexMin([8, 7, 2, 9, 4])

```

```

Logging...
5.0
16.9020µ      1.6901999970286852e-05
... Finished
Logging...
2
12.1130µ      1.2113000025237852e-05
... Finished

```

```

In [352]: 1 def encloseGetPower(f):
2
3           def getPower(*args):
4               return f(*args)
5
6           print(hex(id(getPower)))
7           return getPower
8
9
10 getPower = encloseGetPower(pow)
11 print(getPower(2, 5))
12 print(hex(id(getPower)))
13 print(hex(id(encloseGetPower)))
14 print(getPower.          )
15 print(encloseGetPower.    )

```

```

0x74efc24f9430
32
0x74efc24f9430
0x74efc24f94e0
(<cell at 0x74efc5818ac0: builtin_function_or_method object at 0x74f01fbb0630>,)
None

```

```

In [353]: 1 l = [(1, 2), (-4, -5.5), (10, -10), (-2, 2)]
2          print(sorted(l, key = lambda t: abs(t[0] - t[1]), reverse = True))

```

```

[(10, -10), (-2, 2), (-4, -5.5), (1, 2)]

```

```

In [354]: 1 l = ["s", "0", "1"]
2          card = "".join(reversed(l))
3          print(card)
4          card = ""
5          for k in reversed(l):

```

```
6     card += k
7     print(card)
```

```
10s
10s
```

In [355]:

```
1
2
3
4
5 suits = 'shdc' # Spades, Hearts, Diamonds, Clubs
6 ranks = list('23456789') + ['10', 'J', 'Q', 'K', 'A']
7 cards = [[rank + suit for rank in ranks] for suit in suits]
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```

```
[['As', 'Ks', 'Qs', 'Js', '10s', '9s', '8s', '7s', '6s', '5s', '4s', '3s', '2s'], ['Ah', 'Kh', 'Qh', 'Jh', '10h', '9h', '8h', '7h', '6h', '5h', '4h', '3h', '2h'], ['Ad', 'Kd', 'Qd', 'Jd', '10d', '9d', '8d', '7d', '6d', '5d', '4d', '3d', '2d'], ['Ac', 'Kc', 'Qc', 'Jc', '10c', '9c', '8c', '7c', '6c', '5c', '4c', '3c', '2c']]
[['2s', '3s', '4s', '5s', '6s', '7s', '8s', '9s', '10s', 'Js', 'Qs', 'Ks', 'As'], ['2h', '3h', '4h', '5h', '6h', '7h', '8h', '9h', '10h', 'Jh', 'Qh', 'Kh', 'Ah'], ['2d', '3d', '4d', '5d', '6d', '7d', '8d', '9d', '10d', 'Jd', 'Qd', 'Kd', 'Ad'], ['2c', '3c', '4c', '5c', '6c', '7c', '8c', '9c', '10c', 'Jc', 'Qc', 'Kc', 'Ac']]
```

In [356]:

```
1 data = {'S1': (100, 200, 80, 180), 'S2': (10, 20, 8, 18), 'S3': (50, 150, 50, 150)}
```

```

2
3 def findMinima(data, reverse = False):
4     minimax = min if reverse else max
5     return minimax(data, key = lambda d: data[d][1] - data[d][2])
6
7
8
9
10 print(findMinima(data, reverse = True))
11 print(findMinima(data))
12

```

S2  
S1

In [357]:

```

1 quotes = [('AACC', 6.05, 6.07, 6.03, 6.05, 65800), ('AAME', 1.7, 1.82, 1.7,
2 1.82, 4300), ('AAON', 24.98, 25.07, 24.9, 24.94, 28200), ('AAPL', 317.99,
3 319.57, 316.75, 317.13, 12901800), ('AATI', 3.82, 3.82, 3.74, 3.79, 194600),
4 ('AAWW', 60.89, 61.44, 60.5, 61.19, 272800), ('AAXJ', 65.4, 65.71, 65.28,
5 65.56, 390300), ('ABAT', 4.01, 4.01, 3.95, 3.99, 656300), ('ABAX', 25.26,
6 25.49, 25.04, 25.42, 73700), ('ABBC', 11.75, 11.88, 11.48, 11.53, 29700),
7 ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCD', 3.25, 3.25, 3.11, 3.22,
8 122800), ('ABCO', 48.75, 50.41, 46.9, 50.37, 66300), ('ABCW', 0.52, 0.61, 0.52,
9 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABIO', 3.96, 4,
10 3.88, 4, 38500), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82,
11 0.84, 0.82, 0.83, 28700), ('ABVA', 3.09, 3.25, 3.09, 3.25, 6200), ('ACAD',
12 0.76, 0.76, 0.7, 0.74, 341500), ('ACAS', 7.52, 7.72, 7.52, 7.66, 5199800),
13 ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1,
14 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5,
15 1.6, 12300), ('ACFN', 3.82, 4, 3.82, 3.98, 53700), ('ACGL', 89.76, 90.14,
16 89.39, 89.92, 240900), ('ACGY', 22.41, 22.56, 22.25, 22.46, 86800), ('ACHN',
17 3.12, 3.2, 3.07, 3.16, 113700), ('ACIW', 26.96, 27.03, 26.63, 26.8, 157000),
18 ('ACLI', 33.65, 33.77, 33.45, 33.63, 28700), ('ACLS', 2.47, 2.63, 2.46, 2.53,
19 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACOM', 25.2, 26.6, 24.9,
20 26.56, 265300), ('ACOR', 26.67, 27.07, 26.38, 27.04, 1415000), ('ACPW', 1.84,
21 1.89, 1.77, 1.85, 565500), ('ACTG', 27.2, 27.43, 26.86, 27.18, 228800),
22 ('ACTI', 3.25, 3.26, 3.25, 3.26, 148500), ('ACTS', 2.08, 2.09, 2.07, 2.07,
23 130500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ACWI', 46.53, 46.7, 46.32,
24 46.51, 286200), ('ACWX', 44.49, 44.66, 44.36, 44.6, 55500), ('ACXM', 18, 18.07,
25 17.81, 18.01, 289800), ('ADAM', 7.34, 7.49, 7.33, 7.44, 81700), ('ADAT', 0.6,
26 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300),
27 ('ADCT', 12.68, 12.69, 12.66, 12.68, 1660500), ('ADEP', 6.14, 6.14, 4.95, 5.61,
28 71000), ('ADES', 6.2, 6.22, 6, 6.19, 4800), ('ADGF', 4.31, 4.55, 4.31, 4.54,
29 10200)]

```

```
def myFilter(percentage = 10):
    return list(filter(lambda t: abs((t[4] - t[2]) / t[2]) > (percentage / 100), quotes))

print("10:", myFilter(10))
print("5:", myFilter(5))
print("1:", myFilter(1))
```

```
10: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000)]
5: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ADEP', 6.14, 6.14, 4.95, 5.61, 71000)]
1: [('ABBC', 11.75, 11.88, 11.48, 11.53, 29700), ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82, 0.84, 0.82, 0.83, 28700), ('ACAD', 0.76, 0.76, 0.7, 0.74, 341500), ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1, 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ACHN', 3.12, 3.2, 3.07, 3.16, 113700), ('ACLS', 2.47, 2.63, 2.46, 2.53, 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACPW', 1.84, 1.89, 1.77, 1.85, 565500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ADAT', 0.6, 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300), ('ADEP', 6.14, 6.14, 4.95, 5.61, 71000)]
```

```
In [358]: 1 l = [5, 6, -4, 8]
          2 print(min(l, key = abs))
```

```
-4
```

```
In [462]: 1 importlib.reload(logging)
          2 logging.basicConfig(format = "%(asctime)s %(levelname)s %(message)s", datefmt = "%FT%T")
          3 logging.error("This is an error")
```

```
2026-09-18T20:52:36 ERROR This is an error
```

```
In [360]: 1 logger = logging.getLogger( )
          2 logger.setLevel(logging.DEBUG)
          3 logger.error("This is an error message")
          4 logger.debug("This is a debug message")
```

```
2026-09-18T17:28:00b ERROR This is an error message
2026-09-18T17:28:00b DEBUG This is a debug message
```

```
In [361]: 1 def f():
          2     pass
          3
          4
          5 print(f. )
          6 print(type(f. ))
```

```
f
<class 'str'>
```

```
In [362]: 1 def partial(f):
          2
          3     def callF(*pargs, **kwargs):
          4         logger.debug("Calling your function... " + f. )
```

```

5         start = time.perf_counter()
6         result = f(*pargs, **kwargs)
7         end = time.perf_counter()
8         logger.debug("... Function returned.")
9         logger.info(f"Elapsed Time: {toEnggNotation(end - start)}")
10        # print(f"{toEnggNotation(end - start)}")
11        return result
12
13    return callF
14
15
16    def add(a, b):
17        return a + b
18
19
20    def log3(a):
21        return math.log(a, 3)
22
23
24    @partial
25    def log4(a):
26        return math.log(a, 4)
27
28
29    myAdder = partial(add)
30    logger.info("myAdder: " + str(myAdder(2, 3)))
31    log3 = partial(log3)
32    logger.info("log3: " + str(log3(9)))
33    logger.info("log4: " + str(log4(16)))

```

```

2026-09-18T17:28:00b DEBUG Calling your function... add
2026-09-18T17:28:00b DEBUG ... Function returned.
2026-09-18T17:28:00b INFO Elapsed Time: 1.5930µ 1.5930000358821417e-06
2026-09-18T17:28:00b INFO myAdder: 5
2026-09-18T17:28:00b DEBUG Calling your function... log3
2026-09-18T17:28:00b DEBUG ... Function returned.
2026-09-18T17:28:00b INFO Elapsed Time: 20.6290µ 2.0628999948257842e-05
2026-09-18T17:28:00b INFO log3: 2.0
2026-09-18T17:28:00b DEBUG Calling your function... log4
2026-09-18T17:28:00b DEBUG ... Function returned.
2026-09-18T17:28:00b INFO Elapsed Time: 4.6590µ 4.659000012452452e-06
2026-09-18T17:28:00b INFO log4: 2.0

```

In [363]:

```

1    def cache(f):
2        print("Initializing cache...")
3        cacheDict = {}
4
5        def callF(*pargs):
6            if pargs in cacheDict:
7                print("Cache Hit")
8                return cacheDict[pargs]
9            else:
10               print("Cache Miss")
11               result = f(*pargs)

```

```

12             cacheDict[pargs] = result
13             return result
14
15         return callF
16
17
18     @cache
19     def add(a, b):
20         return a + b
21
22
23     @cache
24     def mul(a, b):
25         return a * b

```

Initializing cache...

Initializing cache...

```

In [364]: 1 print(add.      )
          2 print(mul.    )

```

(<cell at 0x74efc252cdf0: dict object at 0x74efc253fb00>, <cell at 0x74f01f194070: function object at 0x74efc2537b60>)

(<cell at 0x74efc252e310: dict object at 0x74efc253fb40>, <cell at 0x74efc252f220: function object at 0x74efc2537d70>)

```

In [365]: 1 print(add(2, 3))
          2 print(mul(2, 3))
          3 print(add(2, 3))
          4 print(mul(2, 3))

```

Cache Miss

5

Cache Miss

6

Cache Hit

5

Cache Hit

6

```

In [366]: 1 print(type(func tools.lru_cache))

```

<class 'function'>

```

In [367]: 1 @func tools.lru_cache(maxsize = 2)
          2 def graham(a, b, c):
          3     print("Cache Miss")
          4     return a ** b ** c
          5
          6
          7 print(graham(2, 2, 4))

```

Cache Miss

65536

```
In [368]: 1 print(graham(2, 2, 4))
          2 print(graham(2, 2, 4))
          3 print(graham(2, 2, 5))
          4 print(graham(2, 2, 6))
          5 print(graham(2, 2, 4))
```

```
65536
65536
Cache Miss
4294967296
Cache Miss
18446744073709551616
Cache Miss
65536
```

```
In [369]: 1 @functools.lru_cache(maxsize = 100)
          2 def fib(n):
          3     print(f"Calling fib{n}...")
          4     if n <= 1:
          5         return n
          6     return fib(n - 1) + fib(n - 2)
          7
          8
          9 print(fib(2000))
```

Calling fib2000...  
Calling fib1999...  
Calling fib1998...  
Calling fib1997...  
Calling fib1996...  
Calling fib1995...  
Calling fib1994...  
Calling fib1993...  
Calling fib1992...  
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9531698287448247299226390183371677806060701161549788671987985831146887087626459736908672288402365442
2295243347964480139515349562972087652656069529806499841977448720155612802665404554171717881930324025
204312082516817125
```

In [370]:

```
1 def myTimer(f):
2
3     def callF(*args):
4         start = time.perf_counter()
5         result = f(*args)
```

```

6         end = time.perf_counter()
7         print(toEnggNotation(end - start))
8         return result
9
10        return callF
11
12
13    # timedPower = myTimer(math.pow)
14    # print(timedPower(5, 188))
15    #
16    #
17    @myTimer
18    def myPower(a, b):
19        return a ** b
20
21
22    print(myPower(5, 188))

```

```

2.5650µ 2.5650000452515087e-06
2548947057811923643246208636428338888945767288308190057718217365417860642420893057508943799552292297
50557351508177816867828369140625

```

```
In [371]: 1 PRECISION = 4
```

```
In [372]: 1 a = decimal.Decimal(math.e)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.718281828459045090795598298427648842334747314453125
<class 'decimal.Decimal'>
2.718281828459045

```

```
In [373]: 1 a = decimal.Decimal("2.404040313893824388847472828")
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.404040313893824388847472828
<class 'decimal.Decimal'>
2.4040403138938244

```

```
In [374]: 1 a = decimal.Decimal(2.2384892948592929275758693)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.23848929485929293292656439007259905338287353515625
<class 'decimal.Decimal'>
2.238489294859293

```

```
In [375]: 1 l = [decimal.Decimal(random.random()) for _ in range(10)]
2 a = sum(l) / len(l)
```

```
3 print(a)
4 print(type(a))
```

```
0.6355070068397963400919081777
<class 'decimal.Decimal'>
```

In [376]:

```
1 def normalize(f):
2
3     def callF(*args):
4         result = f(*args)
5         # print(type(result))
6         result = float(result)
7         return float(f"{result:.{PRECISION}f}")
8
9     return callF
```

In [377]:

```
1 @normalize
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

In [378]:

```
1 perc_diff(5088, 2872)
```

Out[378]: -43.5535

In [379]:

```
1 avg(2.4, 4.1, 5.932929, 8.2838485)
```

Out[379]: 5.1792

In [380]:

```
1 def add(x, y):
2     time.sleep(2)
3     return x + y
4
```

```
5 print(timeit.timeit("add(2340320832580385043843805805,  
6 320342408320482804320584305)", globals = globals(), number = 10))
```

20.000953659999993

In [381]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.timeit("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

2.0001189919999547

In [382]:

```
1 def add(x, y):  
2     time.sleep(2)  
3     return x + y  
4  
5  
6 print(timeit.repeat("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

[20.000912009000047, 20.000890174000006, 20.000836774999982, 20.00079347799999, 20.000815604000024]

In [383]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.repeat("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

[2.000094929999989, 1.943999961895315e-06, 1.5530000041508174e-06, 1.4529999816659256e-06,  
1.4830000054644188e-06]

In [384]:

```
1 def graham(x, y, z):  
2     return x ** y ** z  
3  
4  
5 print(toEnggNotation(timeit.timeit("graham(4, 5, 6)", globals = globals(),  
number = 10)))
```

686.5070μ 0.0006865070000117157

In [385]:

```
1 @functools.lru_cache  
2 def graham(x, y, z):  
3     return x ** y ** z  
4  
5  
6 print(toEnggNotation(timeit.timeit("graham(2, 3, 4)", globals = globals(),  
number = 10)))
```

3.9570μ 3.9569999898958486e-06

```
In [386]: 1 print(toEnggNotation(timeit.timeit("2 ** 3 ** 4", number = 100)))
```

18.8660μ 1.8866000004891248e-05

```
In [387]: 1 def normalize(precision):
2
3     def callF(f):
4
5         def callFInner(*args):
6             result = f(*args)
7             result = f"{result:.{precision}f}"
8             return result
9
10        return callFInner
11
12    return callF
```

```
In [388]: 1 @normalize(2)
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize(4)
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize(8)
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

```
In [389]: 1 print(perc_diff(3, 7))
2 print(sum_squares(3, 4, 5))
3 print(avg(1.25, 1.75, 1.5, 1.0, 2.0, 2.5, 2.25, 2.75))
```

133.33  
50.0000  
1.87500000

```
In [390]: 1 def normalize(f):
2
3     def callF(*args):
```

```
4         result = f(*args)
5         return result
6
7     return callF
```

In [391]:

```
1  try:
2
3      @normalize(2)
4      def perc_diff(x, y):
5          try:
6              return (y - x) / x * 100
7          except ZeroDivisionError:
8              return 0
9
10     @normalize(4)
11     def sum_squares(*args):
12         return sum(x ** 2 for x in args)
13
14     @normalize(8)
15     def avg(*args):
16         if len(args) == 0:
17             return 0
18
19         numbers = [decimal.Decimal(x) for x in args]
20         sum_ = sum(numbers)
21         return sum_ / len(numbers)
22
23 except TypeError as ex:
24     print(type(ex).      + ": " + ex.args[0] + ".")
```

TypeError: 'int' object is not callable.

In [392]:

```
@functools.lru_cache
1 def getFibonacci(i):
2     if i == 0:
3         return 1
4     if i == 1:
5         return 1
6     if i < 0:
7         return 0
8     else:
9         return getFibonacci(i - 1) + getFibonacci(i - 2)
10
11 print(getFibonacci(60))
12
13 with open("../var/out000.txt", "w") as fl:
14     fl.write("i,fib,fact,gcd\n")
15     for i in range(101):
16         fl.write(f"{i},{math.factorial(i)},{getFibonacci(i)},
17 {math.gcd(getFibonacci(i),math.factorial(i))}\n")
18
19 0
```

```
1 with open("../var/out000.txt") as fl:
1   for line in fl:
1     print(line.strip())
2
1
3
1
4
1
5
1
6
1
7
1
8
1
9
2
0
2
1
2
2
```

2504730781961  
i,fib,fact,gcd  
0,1,1,1  
1,1,1,1  
2,2,2,2  
3,6,3,3  
4,24,5,1  
5,120,8,8  
6,720,13,1  
7,5040,21,21  
8,40320,34,2  
9,362880,55,5  
10,3628800,89,1  
11,39916800,144,144  
12,479001600,233,1  
13,6227020800,377,13  
14,87178291200,610,10  
15,1307674368000,987,21  
16,20922789888000,1597,1  
17,355687428096000,2584,136  
18,6402373705728000,4181,1  
19,121645100408832000,6765,165  
20,2432902008176640000,10946,26  
21,51090942171709440000,17711,1  
22,112400072777607680000,28657,1  
23,25852016738884976640000,46368,46368  
24,620448401733239439360000,75025,25  
25,1551121004330985984000000,121393,1  
26,403291461126605635584000000,196418,34  
27,10888869450418352160768000000,317811,39  
28,304888344611713860501504000000,514229,1  
29,8841761993739701954543616000000,832040,440  
30,265252859812191058636308480000000,1346269,1  
31,8222838654177922817725562880000000,2178309,21  
32,263130836933693530167218012160000000,3524578,2  
33,8683317618811886495518194401280000000,5702887,1  
34,295232799039604140847618609643520000000,9227465,65  
35,10333147966386144929666651337523200000000,14930352,139536  
36,371993326789901217467999448150835200000000,24157817,1  
37,13763753091226345046315979581580902400000000,39088169,37  
38,52302261746660111760007224100074291200000000,63245986,2  
39,20397882081197443358640281739902897356800000000,102334155,1155  
40,815915283247897734345611269596115894272000000000,165580141,1  
41,33452526613163807108170062053440751665152000000000,267914296,3016  
42,1405006117752879898543142606244511569936384000000000,433494437,1  
43,60415263063373835637355132068513997507264512000000000,701408733,129  
44,2658271574788448768043625811014615890319638528000000000,1134903170,170  
45,119622220865480194561963161495657715064383733760000000000,1836311903,1  
46,5502622159812088949850305428800254892961651752960000000000,2971215073,1  
47,258623241511168180642964355153611979969197632389120000000000,4807526976,4358592  
48,12413915592536072670862289047373375038521486354677760000000000,7778742049,13  
49,608281864034267560872252163321295376887552831379210240000000000,12586269025,275  
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55,1269640335365827592596510084756651695958032105144943676227584000000000000,225851433717,55419  
56,71099858780486345185404564746372494973649797888116845868744704000000000000,365435296162,74  
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1  
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1,1  
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842,442  
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00,190392490709135,20735  
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00000,498454011879264,44930592  
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94048000000000000000000,61305790721611591,1  
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84,3314240134565353266999387579130131288000666286242049487118846032383059131291716864129885722968716  
7531561779200000000000000000,259695496911122585,5  
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240182751232000000000000000000,420196140727489673,1  
86,2422709538367273238176552320344125971528487055242938175083876449672016224974245027678946463490131  
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7935047344717824000000000000000000,1100087778366101931,903  
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018284166335168512000000000000000000,1779979416004714189,1  
89,1650795516090846108121691926245361930983966623649654185491352070783317103437850973939991257078760

```

066272908038299975680000000000000000000,2880067194370816120,268748920
90,1485715964481761497309522733620825737885569961284688766942216863704985393094065876545992131370884
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92,1243841405464130725547532432587355307757799171587541435684023958293813771098351951844304612383704
134735310748698265675366400000000000000000000,12200160415121876738,2
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26512612841896796781676470678323200000000000000000000,135301852344706746049,36569
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1828625369792082722375825118521091686400000000000000000000,573147844013817084101,1

```

In [393]:

```

fibonacciNumbers = {}
1 factorialNumbers = {}
gcd = {}
2
with open("../var/out000.txt") as fl:
3     next(fl)
    for ln in fl:
4         l = ln.strip().split(",")
        factorialNumbers[int(l[0])] = int(l[1])
5         fibonacciNumbers[int(l[0])] = int(l[2])
        gcd[(int(l[1]), int(l[2]))] = int(l[3])
6
7 # @functools.lru_cache
def getFibonacci(i):
8     print(f"getFibonacci({i}):")
    if i < 0:
9         return 0
    if i in fibonacciNumbers:
1        print("Found in cache!")
        return fibonacciNumbers[i]
1        print("Not found in cache...")
    fibonacciNumbers[i] = getFibonacci(i - 1) + getFibonacci(i - 2)
2        return fibonacciNumbers[i]
1
3
1 print(getFibonacci(110))
4 print(getFibonacci(11))
1
5
1 def f(i):

```

```

6
1      def g():
7          return math.factorial(i)
1
8      return g
1
9
2  def getFactorial(i):
0      print(f"getFactorial({i}):")
2      if i in factorialNumbers:
1          print("Found in cache!")
2          return factorialNumbers[i]
2      print("Not found in cache...")
2      return f(i)()
3          # return factorialNumbers.get(i, f(i)())      # Doesn't work because
2          the closure is always called. Have to use condition.
4
2
5  print(getFactorial(110))
2  print(getFactorial(11))
6
2
7  def f1(i, j):
2
8      def g1():
2          return math.gcd(i, j)
9
3      return g1
0
3
1  def getGCD(i, j):
3      print(f"getGCD({i, j}):")
2      if (i, j) in gcd:
3          print("Found in cache!")
3          return gcd[(i, j)]
3      print("Not found in cache...")
4      return math.gcd(i, j)
3
5
3  print(getGCD(5040, 21))
6  print(getGCD(5040, 22))
3
7
3
8
3
9
4
0
4
1
4

```

2  
4  
3  
4  
4  
4  
5  
4  
6  
4  
7  
4  
8  
4  
9  
5  
0  
5  
1  
5  
2  
5  
3  
5  
4  
5  
5  
5  
6  
5  
7  
5  
8  
5  
9  
6  
0  
6  
1  
6  
2  
6  
3  
6  
4  
6  
5  
6  
6  
6  
7  
6

8  
6  
9  
7  
0  
7  
1

[illegible]

```
In [394]: 1 def importMPL():
2         import matplotlib as mpl
3
4         return mpl
5
6
7 mpl = importMPL()
8 print(dir(mpl))
```

```
['ExecutableNotFoundError', 'MatplotlibDeprecationWarning', 'MutableMapping', 'Parameter', 'Path', 'RcParams', '_ExecInfo', '_VersionInfo', '__all__', '__bibtex__', '__builtins__', '__cached__', '__doc__', '__file__', '__getattr__', '__loader__', '__name__', '__package__', '__path__', '__spec__', '_add_data_doc', '_api', '_c_internal_utils', '_check_versions', '_cm', '_cm_bivar', '_cm_listed', '_cm_multivar', '_color_data', '_deprecated_ignore_map', '_deprecated_map', '_deprecated_remain_as_none', '_docstring', '_ensure_handler', '_enums', '_fontconfig_pattern', '_get_config_or_cache_dir', '_get_executable_info', '_get_ssl_context', '_get_version', '_get_xdg_cache_dir', '_get_xdg_config_dir', '_image', '_init_tests', '_label_from_arg', '_log', '_logged_cached', '_open_file_or_url', '_parse_to_version_info', '_path', '_preprocess_data', '_rc_params_in_file', '_replacer', '_val_or_rc', '_version', 'artist', 'atexit', 'backends', 'bezier', 'bivar_colormaps', 'cbook', 'cm', 'color_sequences', 'colorizer', 'colormaps', 'colors', 'contextlib', 'cycler', 'defaultParams', 'ft2font', 'functools', 'get_backend', 'get_cachedir', 'get_configdir', 'get_data_path', 'importlib', 'inspect', 'interactive', 'is_interactive', 'locale', 'logging', 'matplotlib_fname', 'multivar_colormaps', 'namedtuple', 'os', 'parse_version', 'path', 'pprint', 'rc', 'rcParams', 'rcParamsDefault', 'rcParamsOrig', 'rc_context', 'rc_file', 'rc_file_defaults', 'rc_params', 'rc_params_from_file', 'rcdefaults', 'rcsetup', 're', 'sanitize_sequence', 'scale', 'set_loglevel', 'shutil', 'subprocess', 'sys', 'tempfile', 'ticker', 'transforms', 'use', 'validate_backend']
```

```
In [395]: 1 epochTime = time.gmtime(0)
2         print(epochTime.tm_year, epochTime.tm_mday, epochTime.tm_hour,
epochTime.tm_min, epochTime.tm_zone)
```

```
1970 1 0 0 GMT
```

```
In [396]: 1 time.perf_counter()
```

```
Out[396]: 440.340111055
```

```
In [397]: 1 start = time.perf_counter()
2         time.sleep(3)
3         end = time.perf_counter()
4         print(end - start)
```

```
3.0002507969999783
```

```
In [398]: 1 print(time.strftime("%I:%M %p %z", time.gmtime(time.time())))
```

```
12:00 PM +0000
```

```
In [ ]: 1 print(datetime.datetime.now(datetime.UTC))
2        print(datetime.datetime.now())
```

```
2026-09-18 12:00:08.720941+00:00
2026-09-18 17:30:08.721215
```

```
In [400]: 1 print(time.strftime("%Y-%m-%dT%H:%M:%S%z", time.gmtime(time.time())))
```

2026-09-18T12:00:08+0000

```
In [401]: my_time_struct = time.strptime("2020-09-17T16:36:16Z", "%Y-%m-%dT%H:%M:%S%z")
1 print(my_time_struct.tm_year, my_time_struct.tm_mon, my_time_struct.tm_mday,
2 my_time_struct.tm_hour, my_time_struct.tm_min, my_time_struct.tm_sec,
my_time_struct.tm_gmtoff)
```

2020 9 17 16 36 16 0

```
In [402]: my_time_struct = time.strptime("2020-09-17T16:36:16+0530", "%Y-%m-%dT%H:%M:%S%z")
1 print(my_time_struct.tm_year, my_time_struct.tm_mon, my_time_struct.tm_mday,
2 my_time_struct.tm_hour, my_time_struct.tm_min, my_time_struct.tm_sec,
my_time_struct.tm_gmtoff)
```

2020 9 17 16 36 16 19800

```
In [403]: 1 print("hello")
```

hello

```
In [404]: 1 s = "Monday, April 18, 2020"
2 myTimeStruct = time.strptime(s, "%A, %B %d, %Y")
3 print(myTimeStruct)
```

time.struct\_time(tm\_year=2020, tm\_mon=4, tm\_mday=18, tm\_hour=0, tm\_min=0, tm\_sec=0, tm\_wday=0, tm\_yday=109, tm\_isdst=-1)

```
In [420]: 1 myDate = datetime.date(2020, 8, 20)
2 print(myDate)
3 print(repr(myDate))
4 print(myDate.day, myDate.month)
```

2020-08-20  
datetime.date(2020, 8, 20)  
20 8

```
In [419]: 1 print(myDate.isoformat())
```

2020-08-20

```
In [421]: 1 myDate = datetime.date.today()
2 print(myDate)
3 print(repr(myDate))
4 print(myDate.day, myDate.month)
```

2026-09-18  
datetime.date(2026, 9, 18)  
18 9

```
In [ ]: 1 print(myDate.isoformat())
```

2026-09-18  
18 9

```
In [407]: 1 myDate = datetime.date.fromtimestamp(time.time())
          2 print(myDate.year)
```

2026

```
In [408]: 1 myDate = datetime.date.fromtimestamp(0)
          2 print(myDate.year)
```

1970

```
In [ ]: 1 myDate = datetime.date.fromisoformat("2020-08-24")
        2 print(myDate)
        3 print(repr(myDate))
```

2020-08-24

datetime.date(2020, 8, 24)

```
In [410]: 1 myTime = datetime.time(15, 23, 45)
          2 print(myTime.isoformat())
```

15:23:45

```
In [411]: 1 print(datetime.time(15, 30, 45, 600))
```

15:30:45.000600

```
In [424]: 1 try:
          2     myTime = datetime.time.now()
          3     print(myTime)
          4 except AttributeError as ex:
          5     print(type(ex).__name__, ex.args)
```

AttributeError ("type object 'datetime.time' has no attribute 'now'",)

```
In [412]: 1 print(datetime.time.fromisoformat("12:34:45.123"))
```

12:34:45.123000

```
In [413]: 1 myTime = datetime.time.fromisoformat("12:34:45.1234")
          2 print(myTime)
          3 print(myTime.hour)
          4 print(myTime.microsecond)
          5 try:
          6     print(myTime[:3])
          7 except TypeError as ex:
          8     print(type(ex).__name__, ex.args)
```

12:34:45.123400

12

123400

TypeError ("'datetime.time' object is not subscriptable",)

```
In [423]: 1 try:
          2     myTime = datetime.time.fromtimestamp(0)
          3     print(myTime)
```

```
4 except AttributeError as ex:
5     print(type(ex)., ex.args)
```

AttributeError ("type object 'datetime.time' has no attribute 'fromtimestamp',)

```
In [414]: 1 myDateTime = datetime.datetime(2020, 12, 9, 12, 43, 32, 38829)
2         print(myDateTime.isoformat())
```

2020-12-09T12:43:32.038829

```
In [426]: 1 myDateTime = datetime.datetime.now()
2         print(myDateTime)
3         print(repr(myDateTime))
4         print(myDateTime.hour)
5         print(myDateTime.isoformat())
```

2026-09-18 18:03:08.224995  
datetime.datetime(2026, 9, 18, 18, 3, 8, 224995)  
18  
2026-09-18T18:03:08.224995

```
In [429]: 1 myDateTime = datetime.datetime.fromtimestamp(0)
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

1970-01-01 05:30:00  
5  
1970-01-01T05:30:00

```
In [436]: 1 myDateTime = datetime.datetime.fromtimestamp(time.time())
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

2026-09-18 18:11:43.908956  
18  
2026-09-18T18:11:43.908956

```
In [431]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02T15:23:44+0545")
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

2025-02-02 15:23:44+05:45  
15  
2025-02-02T15:23:44+05:45

```
In [432]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02T15:23:44")
2         print(myDateTime)
3         print(myDateTime.hour)
4         print(myDateTime.isoformat())
```

2025-02-02 15:23:44  
15  
2025-02-02T15:23:44

```
In [433]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02 15:23:44")
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.isoformat())
```

```
2025-02-02 15:23:44
15
2025-02-02T15:23:44
```

```
In [434]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02 15:23")
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.isoformat())
```

```
2025-02-02 15:23:00
15
2025-02-02T15:23:00
```

```
In [435]: 1 myDateTime = datetime.datetime.fromisoformat("2025-02-02")
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.isoformat())
```

```
2025-02-02 00:00:00
0
2025-02-02T00:00:00
```

```
In [438]: 1 myDateTime = datetime.datetime.utcnow()
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.tzinfo)
          5 print(myDateTime.isoformat())
```

```
2026-09-18 12:54:43.306385
12
None
2026-09-18T12:54:43.306385

/tmp/ipykernel_18021/573860576.py:1: DeprecationWarning: datetime.datetime.utcnow() is deprecated
and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in
UTC: datetime.datetime.now(datetime.UTC).
  myDateTime = datetime.datetime.utcnow()
```

```
In [440]: 1 myDateTime = datetime.datetime.now(datetime.UTC)
          2 print(myDateTime)
          3 print(myDateTime.hour)
          4 print(myDateTime.tzinfo)
          5 print(myDateTime.isoformat())
```

```
2026-09-18 12:55:53.446540+00:00
12
UTC
2026-09-18T12:55:53.446540+00:00
```

```
In [442]: 1 try:
          2     myDateTime = datetime.datetime.now(datetime.IST)
          3     print(myDateTime)
```

```

4     print(myDateTime.hour)
5     print(myDateTime.tzinfo)
6     print(myDateTime.isoformat())
7 except AttributeError as ex:
8     print(type(ex).      , ex.args)

```

AttributeError ("module 'datetime' has no attribute 'IST',")

In [415]: 1 print(cmath.sqrt(-4))

2j

In [451]:

```

1 dt1 = datetime.datetime.now()
2 print(dt1)
3 dt2 = datetime.datetime.today()
4 print(dt2)
5 td = dt1 - dt2
6 print(toEnggNotation(abs(td.total_seconds()))))

```

2026-09-18 18:53:23.132215  
 2026-09-18 18:53:23.132384  
 169.00000μ 0.000169

In [459]:

```

1 dt1 = datetime.datetime.now()
2 print(dt1)
3 startOfDay = datetime.datetime.fromisoformat(datetime.date.today().isoformat())
4 print(startOfDay)
5 td = dt1 - startOfDay
6 print(td.total_seconds())

```

2026-09-18 18:57:44.890444  
 2026-09-18 00:00:00  
 68264.890444

In [461]:

```

1 td = datetime.timedelta(weeks = 52, seconds = 86400)
2 print(datetime.datetime.now() + td)

```

2027-09-18 19:01:37.580957

In [467]:

```

1 print(time.strftime("%FT%T", time.strptime("1/6/2024 2:30 PM", "%d/%m/%Y %I:%M%p")))

```

2024-06-01T14:30:00

In [472]:

```

1 td = datetime.timedelta(days = -1, seconds = 5400)
2 print(td + datetime.datetime.now())
3 print(td.total_seconds() // 60 / 60)

```

2026-09-18 18:41:55.925042  
 -22.5

In [480]:

```

1 myTZ = datetime.timezone(datetime.timedelta(hours = 5.5), name = "IST")
2 print(myTZ)
3 UTCTZ = datetime.timezone.utc
4 print(UTCTZ)

```

IST  
UTC

In [489]:

```
1 print(datetime.datetime.now().replace(tzinfo =  
datetime.timezone(datetime.timedelta(hours = 5.5), name =  
"IST")).astimezone(datetime.timezone(datetime.timedelta(hours =  
9))).isoformat())
```

2026-09-19T21:28:11.050467+09:00

In [507]:

```
1 currentTime = datetime.datetime.now()  
2 tzIST = datetime.timezone(datetime.timedelta(hours = 5.5))  
3 currentISTTime = currentTime.replace(tzinfo = tzIST)  
4 tzJST = datetime.timezone(datetime.timedelta(hours = 9))  
5 currentJSTTime = currentISTTime.astimezone(tzJST)  
6 print(currentJSTTime)  
7 currentJSTTime = currentJSTTime.replace(tzinfo = None)  
8 tzDifference = currentJSTTime - datetime.datetime.now()  
9 print(tzDifference)  
10 print(type(tzDifference).      )  
11 print(round(tzDifference.total_seconds() // 60 / 60, 1))
```

2026-09-19 21:49:02.698258+09:00  
3:29:59.999810  
timedelta  
3.5

In [514]:

```
1 epochTime = 12345678.9  
2 dt = datetime.datetime.fromtimestamp(epochTime).replace(day = 1, hour = 0,  
3 minute = 0, second = 0, microsecond = 0)  
print(repr(dt))
```

datetime.datetime(1970, 5, 1, 0, 0)

In [ ]:

```
1 epochTime = 12345678.9  
2 dt = datetime.datetime.fromtimestamp(epochTime)  
3 print(repr(datetime.datetime(year = dt.year, month = dt.month, day = 1)))
```

datetime.datetime(2026, 9, 1, 0, 0)

In [ ]:

```
1 epochTime = time.time()  
2  
3 dt = datetime.datetime.fromtimestamp(epochTime)  
4 print(repr(datetime.datetime(year = dt.year, month = dt.month, day = 1)))
```

datetime.datetime(2026, 9, 1, 0, 0)

In [520]:

```
1 start = "2001-01-01T13:50:23"  
2 end = "2001-06-12T14:23:50"  
3 diff = int(round((datetime.datetime.fromisoformat(end) -  
4 datetime.datetime.fromisoformat(start)).total_seconds() // 60 / 60, 0))  
print(diff)
```

3889

In [544]:

```
1 start = "2001-01-01T00:00:00"
2 end = "2001-01-01T01:00:00"
3 diff = int(round((datetime.datetime.fromisoformat(end) -
4 datetime.datetime.fromisoformat(start)).total_seconds() // 60 / 60, 0))
5 print(diff)
```

1

In [545]:

```
1 start = "2001-01-01T00:00:00"
2 end = "2001-01-02T01:50:00"
3 diff = int(round((datetime.datetime.fromisoformat(end) -
4 datetime.datetime.fromisoformat(start)).total_seconds() // 60 / 60, 0))
5 print(diff)
```

26

In [534]:

```
1 def formatDateTime(dateTimeObj, formatStr = "%FT%T"):
2     return dateTimeObj.strftime(formatStr)
3
4
5 dateTimeObj = datetime.datetime.strptime("02/04/2025 03:30:30 PM", "%m/%d/%Y
6 %I:%M:%S %p")
7 print(formatDateTime(dateTimeObj))
8 print(formatDateTime(dateTimeObj, "%Y-%m-%d"))
```

2025-02-04T15:30:30

2025-02-04

In [546]:

```
1 def formatDateTime(dateTimeObj, formatStr = "%FT%T"):
2     return dateTimeObj.strftime(formatStr)
3
4
5 dateTimeObj = datetime.datetime(2020, 2, 1, 13, 34, 5)
6 print(formatDateTime(dateTimeObj))
7 print(formatDateTime(dateTimeObj, "%Y-%m-%d"))
```

2020-02-01T13:34:05

2020-02-01

In [50]:

```
1 with open("../var/actors.csv") as fh:
2     itr = csv.reader(fh)
3     for row in itr:
4         print(*row)
```

First Name Last Name DOB Sketches

John Cleese 10/27/39 The Cheese Shop, Ministry of Silly Walks, It's the Arts

Eric Idle 3/29/43 The Cheese Shop, Nudge Nudge, "Spam"

Peter O'Toole 8/2/32 Lawrence of Arabia

In [ ]:

```
1 with open("../var/actors.pdv") as fh:
2     itr = csv.reader(fh, delimiter = "|", quotechar = "'", escapechar =
3     "\\ ", skipinitialspace = True)
4     for row in itr:
5         print(row)
```

```
['FIRST_NAME', 'LAST_NAME', 'DOB', 'SKETCHES']  
['John', 'Cleese', '10/27/39', "The Cheese Shop, Ministry of Silly Walks, It's the Arts"]  
['Eric', 'Idle', '3/29/43', 'The Cheese Shop, Nudge Nudge, "Spam"']  
['Peter', "O'Toole", '8/2/32', 'Lawrence of Arabia']
```

```
In [14]: 1 csv.register_dialect("pdv", delimiter = "|", quotechar = "'", skipinitialspace  
2         = True, escapechar = "\\")  
        csv.list_dialects()
```

```
Out[14]: ['excel', 'excel-tab', 'unix', 'mine', 'pdv']
```

```
In [563]: 1 with open("../var/actors.pdv") as fh:  
2         itr = csv.reader(fh, dialect = "pdv")  
3         for row in itr:  
4             print(row)
```

```
['FIRST_NAME', 'LAST_NAME', 'DOB', 'SKETCHES']  
['John', 'Cleese', '10/27/39', "The Cheese Shop, Ministry of Silly Walks, It's the Arts"]  
['Eric', 'Idle', '3/29/43', 'The Cheese Shop, Nudge Nudge, "Spam"']  
['Peter', "O'Toole", '8/2/32', 'Lawrence of Arabia']
```

```
In [13]: 1 csv.register_dialect("mine", delimiter = "|", skipinitialspace = True,  
2         escapechar = "\\ ", doublequote = False)  
        csv.list_dialects()
```

```
Out[13]: ['excel', 'excel-tab', 'unix', 'mine']
```

```
In [56]: # csv.register_dialect("mine", delimiter = "|", skipinitialspace = True,  
         quoting = csv.QUOTE_NONE, escapechar = "\\ ", doublequote = False,  
         lineterminator = "\n")  
1 csv.register_dialect("mine", delimiter = "|", skipinitialspace = True, quoting  
2 = csv.QUOTE_NONE, lineterminator = "\n", escapechar = "\\ ", doublequote =  
3 False, quotechar = None)  
   csv.list_dialects()
```

```
Out[56]: ['excel', 'excel-tab', 'unix', 'mine', 'pdv']
```

In [59]:

```
1 print("Current literal contents...")
2 with open("../var/actors.pdv") as fh1:
3     for ln in fh1:
4         print(ln, end = "")
5
6 print("\n\nAfter converting to my format...")
7 with open("../var/actors.pdv") as fh1:
8     with open("../var/actors.tsv", "w") as fh2:
9         itr1 = csv.reader(fh1, dialect = "pdv")
10        itr2 = csv.writer(fh2, dialect = "mine")
11        for row in itr1:
12            itr2.writerow(row)
13
14 with open("../var/actors.tsv") as fh:
15     for row in fh:
16         print(row, end = "")
```

Current literal contents...

FIRST\_NAME| LAST\_NAME| DOB| SKETCHES

John|Cleese| 10/27/39| 'The Cheese Shop, Ministry of Silly Walks, It\'s the Arts'

Eric| Idle| 3/29/43| 'The Cheese Shop, Nudge Nudge, "Spam"'

Peter| 'O\'Toole'| 8/2/32| Lawrence of Arabia

After converting to my format...

FIRST\_NAME|LAST\_NAME|DOB|SKETCHES

John|Cleese|10/27/39|The Cheese Shop, Ministry of Silly Walks, It's the Arts

Eric|Idle|3/29/43|The Cheese Shop, Nudge Nudge, "Spam"

Peter|O'Toole|8/2/32|Lawrence of Arabia

In [566]:

```
1 with open("../var/file1.csv") as fh:
2     for ln in fh:
3         print(ln, end = "")
```

col1,col2,col3

10,20,30

30,40,50

60,70,80

In [571]:

```
1 with open("../var/file1.csv") as fh:
2     itr = csv.reader(fh)
3     smallest, largest = math.inf, -math.inf
4     for row in itr:
5         for fld in row:
6             try:
7                 n = int(fld)
8             except ValueError:
9                 pass
10            if n < smallest:
11                smallest = n
12            if n > largest:
13                largest = n
14
15 print(smallest, largest)
```

10 80

```
In [572]: 1 with open("../var/file2.csv") as fh:
          2     for ln in fh:
          3         print(ln, end = "")
```

```
1, 3.14, 500
20, 1, -2
-1.1, -2.2, -3.3
```

```
In [588]: 1 print(type(2.0))
          2 print(type(2.0) == float)
          3 print(isinstance(2.0, float))
```

```
<class 'float'>
True
True
```

```
In [609]: 1 a = "500.0"
          2 b = "500"
          3 if "." in a:
          4     print("a is fractional")
          5 if "." in b:
          6     print("b is fractional")
```

```
a is fractional
```

```
In [611]: 1 with open("../var/file2.csv") as fh:
          2     itr = csv.reader(fh, skipinitialspace = True)
          3     smallest, largest = math.inf, -math.inf
          4     for row in itr:
          5         for fld in row:
          6             n = float(fld) if "." in fld else int(fld)
          7             if n < smallest:
          8                 smallest = n
          9             if n > largest:
          10                 largest = n
          11
          12 print(smallest, largest)
```

```
-3.3 500
```

```
In [614]: 1 with open("../var/file3.tsv") as fh:
          2     for ln in fh:
          3         print(ln, end = "")
```

```
col1    col2    col3
10       20       30
40       50       60
100      200      300
```

```
In [623]: 1 with open("../var/file3.tsv") as fh:
          2     itr = csv.reader(fh, skipinitialspace = True, delimiter = "\t")
          3     smallest, largest = math.inf, -math.inf
          4     next(fh)
```

```

5         for row in itr:
6             for fld in row:
7                 n = int(fld)
8                 if n < smallest:
9                     smallest = n
10                if n > largest:
11                    largest = n
12
13    print(smallest, largest)

```

10 300

```

In [627]: 1 with open("../var/file4.csv") as fh:
2           for ln in fh:
3               print(ln, end = "")

```

```

-col1-|-col2-|-col3-
10|20|30
-3.14-|-25-|-100-
---3.14-|20|-30-

```

```

In [633]: 1 a = "--3.3-"
2           print(a.removesuffix("-").removeprefix("-"))

```

-3.3

```

In [ ]: 1 print(a.replace("-", ""))

```

3.3

```

In [642]: 1 with open("../var/file4.csv") as fh:
2           itr = csv.reader(fh, delimiter = "|")
3           smallest, largest = math.inf, -math.inf
4           next(fh)
5           for row in itr:
6               for fld in row:
7                   fld = fld.replace("-", "")
8                   n = float(fld) if "." in fld else int(fld)
9                   if n < smallest:
10                       smallest = n
11                   if n > largest:
12                       largest = n
13
14    print(smallest, largest)

```

3.14 100

```

In [648]: 1 a = "b"
2           print(ord(a) - ord("a"))

```

1

```

In [655]: 1 data = [{'a': '1_a', 'b': '1_b', 'c': '1_c'}, {'c': '2_c', 'd': '2_d'}, {'a':
               '3_a', 'c': '3_c', 'e': '3_e'}]
           with open("../var/out2.csv", "w") as fh:

```

```

2      itr = csv.writer(fh)
      itr.writerow("abcde")
3      for d in data:
          row = ["", "", "", "", ""]
4          for k, v in d.items():
              row[ord(k) - ord("a")] = v
5          itr.writerow(row)

6  with open("../var/out2.csv") as fh:
      for ln in fh:
7          print(ln, end = "")
8
9
10
11
11
11
12
11
13

```

```

a,b,c,d,e
1_a,1_b,1_c,,
,,2_c,2_d,
3_a,,3_c,,3_e

```

```

In [8]: 1  for _ in range(10):
2          print(random.random(), end = " ")
3
4  randList = [random.random() for _ in range(5000000)]
5  print("Min:", toEnggNotation(min(randList)))
6  print("Max:", toEnggNotation(max(randList)))
7  minima = min(randList)
8  print(f"{minima:.200f}".rstrip("0"))
9  dec = decimal.Decimal(minima)
10 print(dec)
11 print(dec.normalize())
12 recurring = 1 / 3
13 print(decimal.Decimal(recurring))
14 print(f"{recurring:.200f}")

```

[illegible]

```
In [9]: 1 print(randList[:5])
```

[0.08172163591676762, 0.6379410884560406, 0.36610121100531756, 0.47189283950477456, 0.7181724643263364]

```
In [ ]: 1 recurring = 1 / 10
        2 print(decimal.Decimal(recurring))
        3 print(f"{recurring:.200f}")
```

[illegible]

```
In [89]: 1 recurring = 2000 / 10000
          2 print(f"{recurring:.200f}")
```

[illegible]

```
In [14]: 1 print(f"{random.random():f}")
2 print(random.random())
3 try:
4     print(f"{random.random():b}")
5 except ValueError as ex:
6     print(type(ex).          + ": " + ex.args[0])
7 print(f"{123:b}")
8 print(sys.float_info)
9 try:
10     print(f"{0.1:b}")
11 except ValueError as ex:
12     print(type(ex).          + ": " + ex.args[0])
13 print(int("0b1001", 2))
14 print(int("1001", 2))
15 try:
16     print(int(1001, 2))
17 except TypeError as ex:
18     print(type(ex).          + ": " + ex.args[0])
```

```
1 print(f"{smallestFloat:.1075f}")
2 print(smallestFloat)
```

5e-324

0.0

None

2720295377534757.0 673366308862969.4 5999141237517758.0 6442052165548231.0 1752782305927253.0

```
abc
  abc
..abc
abc  de
abc  de
    abcde
```

```
1
[2, 4, 3]
```

```
7 print(random.choices({0, 1, 2, 3, 4}, k = 3))
```

```

8 except TypeError as ex:
9     print(type(ex).          + ": " + ex.args[0])
10
11 try:
12     print({4, 3, 2, 1, 0}[2])
13 except TypeError as ex:
14     print(type(ex).          + ": " + ex.args[0])

```

```

TypeError: 'set' object is not subscriptable
TypeError: 'set' object is not subscriptable
TypeError: 'set' object is not subscriptable

<>:12: SyntaxWarning: 'set' object is not subscriptable; perhaps you missed a comma?
<>:12: SyntaxWarning: 'set' object is not subscriptable; perhaps you missed a comma?
/tmp/ipykernel_31787/1848623714.py:12: SyntaxWarning: 'set' object is not subscriptable; perhaps you missed a comma?
print({4, 3, 2, 1, 0}[2])

```

```

In [35]: 1 try:
2         print(random.choice({4, 3, 2, 1}))
3     except TypeError as ex:
4         print(type(ex).          + ": " + ex.args[0])

```

```

TypeError: 'set' object is not subscriptable

```

```

In [ ]: 1 face_values = ['9', '10', 'J', 'Q', 'K', 'A']
2
3
4 def make_sample_space(face_values):
5     return [(e1, e2) for e1 in face_values for e2 in face_values]
6
7
8 print(make_sample_space(face_values))

```

```

[('9', '9'), ('9', '10'), ('9', 'J'), ('9', 'Q'), ('9', 'K'), ('9', 'A'), ('10', '9'), ('10', '10'),
('10', 'J'), ('10', 'Q'), ('10', 'K'), ('10', 'A'), ('J', '9'), ('J', '10'), ('J', 'J'), ('J', 'Q'),
('J', 'K'), ('J', 'A'), ('Q', '9'), ('Q', '10'), ('Q', 'J'), ('Q', 'Q'), ('Q', 'K'), ('Q', 'A'),
('K', '9'), ('K', '10'), ('K', 'J'), ('K', 'Q'), ('K', 'K'), ('K', 'A'), ('A', '9'), ('A', '10'),
('A', 'J'), ('A', 'Q'), ('A', 'K'), ('A', 'A')]

```

```

In [41]: 1 sample_space = make_sample_space(face_values)
2
3
4 def simulate_throws_from_sample_space(sample_space, n):
5     return [random.choice(sample_space) for _ in range(n)]
6
7
8 print(simulate_throws_from_sample_space(sample_space, 10))

```

```

[('9', 'K'), ('Q', '9'), ('Q', 'Q'), ('J', 'A'), ('A', 'A'), ('A', 'J'), ('A', 'K'), ('K', 'A'),
('10', 'Q'), ('A', 'K')]

```

```

In [43]: 1 def simulate_throws(face_values, n):
2         return [tuple(random.choices(face_values, k = 2)) for _ in range(n)]
3

```

```
4
5 print(simulate_throws(face_values, 10))
```

```
[('Q', 'A'), ('K', 'Q'), ('10', '10'), ('10', '9'), ('Q', 'Q'), ('A', 'K'), ('A', 'K'), ('A', 'Q'), ('10', 'J'), ('10', 'Q')]
```

In [ ]:

```
1
2
3 def countOccurrences(simulationFunction, nThrows):
4     input = sample_space if simulationFunction ==
5       simulate_throws_from_sample_space else face_values
6     throws = simulationFunction(input, nThrows)
7     occurrences = dict.fromkeys(face_values, 0)
8     for k, v in throws:
9         occurrences[k] = occurrences.get(k, 0) + 1
10        occurrences[v] = occurrences.get(v, 0) + 1
11    return occurrences
12
13 print(countOccurrences(simulate_throws, 10))
14 print(countOccurrences(simulate_throws_from_sample_space, 10))
15
16
17
18
19
20
```

```
[('K', 'A'), ('J', 'J'), ('Q', '10'), ('Q', '9'), ('K', '10'), ('K', '10'), ('K', 'Q'), ('A', '9'), ('K', '9'), ('10', '10')]
{'9': 3, '10': 5, 'J': 2, 'Q': 3, 'K': 5, 'A': 2}
[('J', 'A'), ('9', '10'), ('10', '9'), ('Q', 'J'), ('9', 'A'), ('10', 'K'), ('K', 'J'), ('A', 'K'), ('10', '9'), ('A', '9')]
{'9': 5, '10': 4, 'J': 3, 'Q': 1, 'K': 3, 'A': 4}
```

In [3]:

```
1 def randomFloat(low, hig):
2     return low + (random.random() * (hig - low))
3
4
5 print(max([randomFloat(2, 4) for _ in range(100_000_000)]))
```

3.9999999851096346

In [7]:

```
1 def randomFloat(low, hig):
2     return low + (random.random() * (hig - low))
3
4
5 print(max([randomFloat(2, 2.0001 + sys.float_info.min) for _ in
6 range(100_000_000)]))
```

2.0000999999999641

```
In [14]: 1 print(sum([0.1] * 10))
2 print(f"{sum([0.1] * 10):.1074f}".rstrip("0").rstrip("."))
```

```
1.0
1
```

```
In [25]: 1 print(math.nextafter(1, 2, steps = 1))
2 print(math.nextafter(1, 2, steps = 2))
3 print(math.nextafter(1, 1 + sys.float_info.min, steps = 1))
4 print(math.nextafter(2, 3, steps = 1))
5 print(math.nextafter(100, 101, steps = 1))
6 print(math.nextafter(100_000, 101_000, steps = 1))
7 print(math.nextafter(0.1, 0.2, steps = 1))
8 print(math.nextafter(0.01, 0.02, steps = 1))
9 print(math.nextafter(0.001, 0.002, steps = 1))
```

```
1.0000000000000002
1.0000000000000004
1.0
2.0000000000000004
100.00000000000001
100000.00000000001
0.10000000000000002
0.010000000000000002
0.0010000000000000002
```

```
In [27]: 1 print(math.comb(3, 2))
2 print(math.perm(3, 2))
```

```
3
6
```

```
In [33]: 1 data = [(0, 1), (1, 2, 3), (1, 3, 5, 7), (1, 1, 2, 3, 5, 8, 13)]
2 print([math.hypot(*vec) for vec in data])
```

```
[1.0, 3.7416573867739413, 9.16515138991168, 16.522711641858304]
```

```
In [48]: data = [61, 35, 99, 100, 75, 94, 88, 14, 21, 39, 53, 25, 87, 84, 81, 55, 86,
1 18, 69, 44, 16, 33, 66, 52, 70, 52, 95, 45, 94, 35, 68, 70, 52, 53, 30, 87, 79,
2 51, 92, 72, 55, 40, 15, 74, 86, 87, 91, 70, 45, 37]
3 print("sorted data:", sorted(data))
4 print("count:", len(data))
5 print("unique count:", len(set(data)))
6 print("min:", min(data))
7 print("max:", max(data))
8 print(f"mean: {statistics.fmean(data):.3f}")
9 print(f"std dev: {statistics.stdev(data):.3f}")
10 print("modes:", statistics.multimode(data))
11 print(f"25th percentile: {statistics.quantiles(data, n = 4)[0]:.3f}")
12 print(f"median: {statistics.median(data):.3f}")
13 print(f"75th percentile: {statistics.quantiles(data, n = 4)[2]:.3f}")
14
```

```
9
1
0
1
1
1
2
```

```
sorted data: [14, 15, 16, 18, 21, 25, 30, 33, 35, 35, 37, 39, 40, 44, 45, 45, 51, 52, 52, 52, 53,
53, 55, 55, 61, 66, 68, 69, 70, 70, 70, 72, 74, 75, 79, 81, 84, 86, 86, 87, 87, 87, 87, 88, 91, 92, 94,
94, 95, 99, 100]
count: 50
unique count: 38
min: 14
max: 100
mean: 60.800
std dev: 25.283
modes: [87, 52, 70]
25th percentile: 39.750
median: 63.500
75th percentile: 86.000
```

In [53]:

```
1 print(statistics.quantiles(range(0, 202, 2), n = 4))
2 print(statistics.quantiles(range(0, 202, 2), n = 4, method = "inclusive"))
3 print(statistics.quantiles(range(0, 202, 2), n = 4, method = "exclusive"))
```

```
[49.0, 100.0, 151.0]
[50.0, 100.0, 150.0]
[49.0, 100.0, 151.0]
```

In [55]:

```
1 print(statistics.quantiles(range(0, 1005, 5), n = 4))
2 print(statistics.quantiles(range(0, 1005, 5), n = 4, method = "inclusive"))
3 print(statistics.quantiles(range(0, 1005, 5), n = 4, method = "exclusive"))
```

```
[247.5, 500.0, 752.5]
[250.0, 500.0, 750.0]
[247.5, 500.0, 752.5]
```